

Portal-Backed Graph Architecture for Digital Product Passports: Semantic Governance and Access Control for Multi-Tiered Supply Chains

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Abstract

The European Union’s Ecodesign for Sustainable Products Regulation (ESPR) mandates Digital Product Passports (DPPs) for an expanding range of product categories. The current architectural blueprint envisions flat-file data payloads hosted at the Economic Operator level, accessed via unique identifiers resolving to distributed endpoints. This paper argues that this model carries systemic vulnerabilities — link rot, business continuity failure, cyber-attack exposure, and intellectual property leakage — that will compromise the ESPR’s circular economy objectives over the decades-long lifespan envisioned for DPP data. We propose a portal-backed, graph-based, network-structured DPP architecture built on semantic web standards (RDF, JSON-

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LD, Named Graphs), in which each individual DPP is a knowledge graph node in a production network, linked to its predecessors via semantic properties. All product data — including data from the Responsible Economic Operator — is stored behind a federation of national portals, one per EU member state, with strictly controlled, branch-level access rights ensuring protection of sensitive data. SHACL serves as a unified data modelling language for both conformity checking and user-interface generation, supported by a public repository of ontologised regulations and a modular SHACL brick library. Evidence from the DPP4EU 2026 conference confirms that the constituent elements of this architecture are independently recognised by the practitioner community as necessary. We propose to integrate them into a coherent whole.

Keywords: Digital Product Passport, Ecodesign for Sustainable Products Regulation, CEN-CENELEC JTC 24, GS1 Digital Link, Semantics, SHACL, DigiPass CSA, data authentication, circular economy, material passports,

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1. List of Abbreviations

Table 1: List of abbreviations used in this paper.

Abbreviation	Full Form
ABAC	Attribute-Based Access Control
AAS	Asset Administration Shell
API	Application Programming Interface
CAS	Chemical Abstracts Service
CLP	Classification, Labelling and Packaging
CEN	European Committee for Standardisation
CENELEC	European Committee for Electrotechnical Standardisation
COST	European Cooperation in Science and Technology
CSA	Coordination and Support Action
DAG	Directed Acyclic Graph
DASH	SHACL Application Dashboard
DIN	Deutsches Institut für Normung (German Institute for Standardisation)
DPP	Digital Product Passport
EC	European Commission
ECHA	European Chemicals Agency
eIDAS	Electronic Identification, Authentication and Trust Services
EN	European Norm
EORI	Economic Operators' Registration and Identification
EPD	Environmental Product Declaration
ERP	Enterprise Resource Planning
ESPR	Ecodesign for Sustainable Products Regulation
EU	European Union
GUI	Graphical User Interface
HTTP	HyperText Transfer Protocol
IDSA	International Data Spaces Association

Table 1: List of abbreviations used in this paper (continued).

Abbreviation	Full Form
IEC	International Electrotechnical Commission
IPI	Intellectual Property [Rights Transfer]
IP	Intellectual Property
IRI	International Resource Identifier
ISO	International Organisation for Standardisation
IT	Information Technology
JSON-LD	JavaScript Object Notation for Linked Data
JTC	Joint Technical Committee (CEN/CENELEC JTC 24; ISO/IEC JTC 5)
LCA	Life Cycle Assessment
NGO	Non-Governmental Organization
NIST	National Institute of Standards and Technology
ODRL	Open Digital Rights Language
OECD	Organisation for Economic Co-operation and Development
OWL	Web Ontology Language
PDF	Portable Document Format
PubChem	Open Chemistry Database (U.S. National Library of Medicine)
QR	Quick Response
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
REO	Responsible Economic Operator
REOs	Responsible Economic Operators
REST	Representational State Transfer
RDF	Resource Description Framework
SHACL	Shapes Constraint Language
SME	Small and Medium-sized Enterprise
SPOD	Single Point of Definition

Table 1: List of abbreviations used in this paper (continued).

Abbreviation	Full Form
SPARQL	SPARQL Protocol and RDF Query Language
SVHC	Substances of Very High Concern
TTL	Turtle (RDF serialisation format)
UN/CEFACT	United Nations Centre for Trade Facilitation and Electronic Business
URI	Uniform Resource Identifier
VAT	Value Added Tax
VIES	VAT Information Exchange System
W3C	World Wide Web Consortium
XML	Extensible Markup Language

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- 1 The IRI as a language-neutral identifier. The same concept (“exit”) expressed in multiple natural languages maps to a single International Resource Identifier (IRI). This principle, applied throughout the DPP architecture, enables cross-lingual, cross-jurisdictional interoperability without requiring a single natural language to dominate the system. The terminology is taken from ontologies, which provide the semantic vocabulary for the DPP knowledge graphs. 23
- 2 The production network in terms of products and processes. Each process (A–F) produces a product, and products flow between processes as inputs and outputs. The network is a directed acyclic graph, not a linear chain: products are assembled from multiple inputs, and a single product may be used in multiple downstream processes. Each product flow has an associated DPP. 24
- 3 The corresponding DPP graph network. Each product in the production network has its own DPP, stored as a knowledge graph behind the portal. The DPPs are linked via `dpp:hasPrecursor` (assembly links) and `dpp:derivedFrom` (transformation links). The supply chain information is extracted by traversing the linked DPPs within the portal’s triple-store — not by embedding it in a single monolithic record. The portal controller maintains and updates the link network as the production network evolves. 26
- 4 Conceptual overview of the portal-backed DPP architecture. The federated repository (top) stores DPP data, ontologised regulations, delegated acts, norms, and supervisory information. Professional organisations authorise and license SHACL bricks, which are assembled into schemas for tailored app generation. The federated controller and portal (centre) govern access, modification rights, and updates. Licensed DPP businesses construct tailored applications from the SHACL schemas, ensuring that user-entered data conforms to regulations by construction. The “reader” person app provides public access to DPP data. 29

2. Introduction

2.1. The ESPR Mandate and Its Architectural Implications

The Ecodesign for Sustainable Products Regulation (ESPR), which entered into force in July 2024, establishes the Digital Product Passport as *“the cornerstone of the Commission’s approach to more environmentally sustainable and circular products.”* The regulation requires that DPPs carry product-specific data across the entire lifecycle — from material composition and origin, through manufacturing and use, to repair, recycling, and disposal — accessible to consumers, professional users, circular economy operators, and market surveillance authorities.

The European Commission has set February 2027 as the deadline for the first implementing acts, beginning with a central registry:

“Developing the infrastructure — the IT infrastructure needed in order to make this happen in February 2027. And here, the first piece of legislation that we will adopt is the implementing act for the central registry.”

The first product-specific DPP requirements target batteries under the separate Battery Regulation, with subsequent product categories — textiles, construction products, electronics, chemicals, and others — following progressively.

This paper is motivated by a simple but consequential observation: **the architectural decisions being made now will determine whether DPPs function as durable, trustworthy infrastructure for decades, or whether they degrade into a fragmented landscape of broken links, inaccessible data, and compromised intellectual property.**

2.2. The Current Architectural Direction

The current DPP architectural blueprint, as developed through CEN & CENELEC JTC 24 and articulated in the eight standards recently approved, follows a distributed model:

1. Each Economic Operator (EO) hosts DPP data at their own endpoint.
2. A central registry maps unique product identifiers to EO endpoints.
3. Data is delivered as structured files (currently envisioned as PDF-based or semi-structured payloads).

4. Access is binary — either the endpoint is reachable, or it is not.

As a representative of the German quality infrastructure noted at the DPP4EU 2026 conference [7], the current delivery format remains problematic: *“this continuity that those data which needed in the processing are delivered as a PDF they don’t really help. So we want to have them databased.”* The aspiration for semantic, machine-readable standards persists: “the hope is obviously that it’s going to be a thing in the future and we’re not going to rely on non-semantic representations like PDF.”

2.3. The Proposed Alternative

In contrast to the distributed, flat-file model, this paper proposes a **portal-backed, network-structured DPP architecture** built on semantic web standards. In this model, DPPs are stored as linked graphs within a federation of per-member-state triple-stores, each acting as a semantic proxy archive, rather than as flat files at EO endpoints. Individual DPPs reference their component predecessors via semantic properties (`dpp:hasPrecursor`, `dpp:derivedFrom`), forming a network of linked passports rather than a single monolithic supply chain record. Access governance is enforced through Attribute-Based Access Control (ABAC) with branch-level ODRL policies, ensuring that different user categories — from public consumers to market surveillance authorities — see only the data branches they are authorised to access.

2.4. The Gap This Paper Addresses

While individual components of a robust DPP architecture — semantic representations, access control, data persistency, value-chain traceability — are independently recognised as necessary by the practitioner community, no single proposal integrates them into a coherent architectural whole. This paper fills that gap by proposing an architecture that unifies these elements under a single governance framework.

The necessity of such integration is evident in the tensions visible at the DPP4EU 2026 conference. For example, a representative from the battery passport context described an access control vision that is almost identical to the ABAC model proposed in this paper [8]:

“The DPP then, of course, needs to have those data in public or controlled data. The actor needs to be authenticated and authorised and needs to have those read access rights. So our expected

result then is again that the actor sees those data that he's allowed to see but nothing more."

Yet this access control vision is not connected to a semantic graph architecture or a portal-backed storage model in any existing proposal. Similarly, the need for decades-long data persistency was directly raised at the conference [7]: *"How do you store these data for decades and make them accessible, for example, to recyclers?"* But no presenter at the conference offered a concrete architectural solution to the link-rot and business-continuity risks that distributed REO-hosted storage creates over such time spans.

This paper proposes that the portal-backed graph architecture, originally presented as an honorary lecture at the DPP4EU 2026 conference [8], addresses these gaps in an integrated manner.

2.5. Contributions

This paper makes the following contributions:

1. A formal critique of the distributed flat-file DPP model, identifying four categories of systemic risk: link rot, business continuity failure, cyberattack exposure, and IP leakage.
2. The Networked Individual DPP model, in which supply chains are represented as graphs of semantically linked individual DPPs rather than as monolithic supply chain records embedded in a single passport.
3. A portal controller architecture with two governance loops: an Ingestion Governance Loop (SHACL validation, cryptographic alignment, external entity normalisation) and a Resolution Governance Loop (graph slicing, dynamic SPARQL CONSTRUCT queries based on ABAC policies).
4. An ABAC framework with five standardised user categories:
 - `dpp:PublicUser`
 - `dpp:ProfessionalUser`
 - `dpp:CircularOperator`
 - `dpp:MarketSurveillanceAuthority`
 - `dpp:InternalDataOriginator`

and ODRL-encoded branch-level access policies.

5. A trust tier matrix for entity vetting, combining legal existence verification (EORI / VIES), competence vetting (W3C Verifiable Credentials), and identity binding (eIDAS 2.0).
6. A cross-registry linking strategy for chemical data governance, connecting DPPs to authoritative public databases (ECHA, PubChem) rather than duplicating chemical hazard data.
7. SHACL as a unified data modelling language, serving both conformity checking and GUI generation, supported by a modular SHACL brick library, a two-stage design toolchain, and a public repository for ontologised regulations and standards.

2.6. Paper Structure

Section 2 presents the problem statement and critique of the current distributed flat-file model. Section 3 introduces the Networked Individual DPP model, in which each DPP is a node in a knowledge graph within a production network. Section 4 details the portal architecture, controller governance, the Ingestion and Resolution Governance Loops, and the portal API. Section 5 presents the ABAC framework with five standardised user categories, ODRL branch-level access policies, the trust tier matrix, and the EC Authorisation App. Section 6 presents SHACL as a unified data modelling language, including the SHACL brick library, the two-stage design toolchain, and the public repository for ontologised regulations and standards. Section 7 addresses chemical data governance and cross-registry linking to authoritative external databases. Section 8 discusses implementation considerations, including the role of DigiPass Europe as community sustainer, the SME challenge, cross-border interoperability, and open challenges. Section 9 concludes with a roadmap for the European Commission. Appendix A specifies the portal REST API, and Appendix B provides example API interaction sequences.

3. Problem Statement: Systemic Risks in the Distributed Flat-File DPP Model

3.1. Link Rot and Data Longevity

The ESPR envisions DPP data remaining accessible for the entire product lifecycle — potentially decades. In the distributed model, DPP data is hosted at REO endpoints. When an REO ceases operations, restructures, rebrands, or simply reorganises its IT infrastructure, the DPP endpoint may become unreachable. The central registry tracks the identifier-to-endpoint mapping,

but it does not store the data itself. The DPP4EU 2026 conference explicitly surfaced this concern, with a participant asking “*how do you store these data for decades and make them accessible, for example, to recyclers?*” [10]. The distributed model offers no structural solution. Each REO endpoint is a single point of failure for the DPPs it hosts. Over decades, across millions of products and hundreds of thousands of REOs, the cumulative effect is a progressive decay of accessible DPP data — **link rot at ecosystem scale**.

3.2. Business Continuity

Related to link rot is the broader issue of business continuity. When a company is acquired, dissolved, or restructured, its DPP endpoints may not be maintained. The data — composition information, hazard data, supply chain provenance — does not transfer automatically to a successor entity. For circular economy operators attempting to recycle a product 15 years after manufacture, the DPP may simply not exist anymore. A conference participant from the Materials Commons initiative acknowledged this gap, noting that while the initiative provides “*infrastructure and governance of a federated software system which is capable of storing this kind of data and making it accessible,*” what Materials Commons does not do is “*run servers in the end*” [24]. This reveals a structural tension: the community recognises the need for governed, persistent storage, but the federated approach explicitly stops short of providing it. The gap between recognising the need for persistent storage and actually providing it is precisely the gap that the portal-backed architecture fills.

3.3. Cyberattack Exposure

In the distributed model, each REO endpoint is a potential attack surface. DPP endpoints, by design, must be publicly resolvable (at least for public data branches). This makes them visible targets for denial-of-service attacks, data tampering, and supply chain poisoning attacks. A compromised REO endpoint could serve falsified DPP data — incorrect composition declarations, fabricated certificates, or altered hazard classifications. The scale of the attack surface is proportional to the number of REOs — potentially hundreds of thousands across the European single market. Centralised monitoring and incident response across such a distributed landscape is structurally difficult. Conference participants also raised concerns about the integrity of data carriers, with one speaker noting the need to “*make sure that whatever the code is it goes to the correct destination*” and that “*there*

is a method to detect that there is a clone” [12]. The emphasis on offline and online verification of data carriers reflects an awareness that the current distributed architecture creates verification challenges that a federated portal with unified cryptographic validation could address more robustly.

3.4. Intellectual Property Leakage

The ESPR mandates that DPPs carry composition and supply chain data. However, much of this data is commercially sensitive — formulation details, supplier identities, process parameters. Industry resistance to sharing proprietary composition data through a system that exposes it to competitors is well documented. The distributed flat-file model does not inherently provide branch-level access control. An REO endpoint either serves the complete DPP payload or it does not. Granular, attribute-based filtering of data branches based on the requester’s identity and role is not a native feature of the current architectural blueprint. Conference participants raised exactly this concern, asking *”who indeed will have the access for the full information and data and how it could be managed”* [11]. The JTC 24 standards attempt to address this by providing an environment for access rights management, but the standards only define the framework — the actual enforcement mechanism at the data layer remains unresolved [20].

3.5. The Media Break Problem: From PDF to Machine-Readable Data

A further systemic risk in the current model is what standards practitioners describe as “media breaks” — the loss of semantic content when data transitions between representation formats. As a JTC 24 representative noted at the conference: *“our API are semi-formal but for implementation need full formality so this means our documents are having media breaks”* [23]. The current reliance on PDF as a delivery format for standards and DPP data compounds this problem. As a participant from the German quality infrastructure stated, data *“delivered as a PDF they don’t really help. So we want to have them databased”* [18]. This media break between the semantic aspirations of the DPP ecosystem and the non-semantic reality of current data delivery creates a structural gap that the portal-backed graph architecture addresses by storing DPP data natively as RDF graphs within a triple-store, eliminating the format conversion problem entirely.

3.6. The Machine-Readable Regulation Gap

A deeper structural problem emerged in the conference discussions: the DPP alone is insufficient without machine-readable regulation against which DPP data can be checked. As a JTC 24 representative observed: *"the machine-readable product data must become checkable by machine-readable regulation and legal text. This is future music"* [21]. Without this capability, *"you still have to look in the book whether what you see on the screen is actually according to requirements"* [22]. This observation highlights that the distributed flat-file model addresses only half of the equation — making product data machine-readable — while leaving the regulatory checking side unresolved. A portal-backed architecture with a triple store could, in principle, host both DPP graphs and machine-readable regulatory shapes (expressed, for example, as SHACL constraints), enabling automated compliance checking within a single semantic framework.

3.7. Summary of Systemic Risks

The evidence from the DPP4EU 2026 conference transcripts demonstrates that these risks are not merely theoretical concerns identified by this paper. They are actively discussed, debated, and left unresolved by the practitioner community. The following sections present an architectural response to each of these risks.

4. The Networked Individual DPP Model

4.1. From Monolithic Supply Chain Records to Linked Passports

A fundamental design choice in DPP architecture is whether to represent a product's entire supply chain as a single monolithic record or as a network of linked individual passports. The current flat-file approach implicitly favours the monolithic model: a single DPP document attempts to encode all upstream component data, material origins, and process information within a single payload. This approach has several structural limitations. First, **data staleness**: component data embedded at assembly time becomes stale if the component's DPP is updated later (e.g., a corrected hazard classification or a supplier change). Second, **redundancy and inconsistency**: the same component data may be duplicated across multiple parent product DPPs, leading to inconsistency when updates occur. Third, **size and complexity**: for products with deep, multi-tiered supply chains (e.g., a battery

Table 2: Comparison of systemic risks across DPP architectural models.

Risk Category	Distributed Model	Flat-File	Portal-Backed Model (Proposed)	Graph
Link rot	Each REO is a single point of failure; data decays over time		Portal stores data persistently; no dependency on REO availability	
Business continuity	DPP data lost when REO ceases operations		DPP data persists in portal; independent of REO lifecycle	
Cyberattack exposure	Hundreds of thousands of endpoints, each a target		Federated portal with unified security posture	
IP leakage	Binary access — full payload or nothing		Branch-level ABAC with ODRL policies; selective visibility	
Media breaks	PDF delivery loses semantic content		Native RDF storage eliminates format conversion	
Regulatory checking	Manual interpretation required; “future music”		SHACL shapes enable automated compliance checking	

electric vehicle with thousands of components), a single monolithic DPP becomes unwieldy. Fourth, **modularity breakdown**: as a representative of JTC 24 noted at the DPP4EU 2026 conference [7], once a modular data model reaches hundreds or thousands of data points, dependencies between modules become extremely hard to track — “*you don’t really know what is going to happen to the whole ecosystem*” if a single referenced data point is modified.

The Networked Individual DPP model proposes an alternative: each product, component, and material batch receives its own DPP, and supply chain relationships are represented as **semantic links** between these individual passports. A parent product’s DPP does not embed its components’ data; it simply references their DPP URIs. This approach mirrors the structure already recognised in the standards community. As a presenter from the University of Pavia described at DPP4EU 2026 [7], information units

— documents, top-level documents, sections, and parts of sections — are “*all represented uniformly as information units which can cite one another. An information unit can have one parent at most. This captures the hierarchical structure of standards and information units can be derived from one or more parent or one or more source information units*”. This description — hierarchical structure, mutual citation, derivation from parent units — is structurally identical to the semantic linking model proposed here for DPPs.

4.2. Core Semantic Linking Properties

The Networked Individual DPP model is built on three core semantic properties, expressed in RDF/RDFS vocabulary consistent with the architecture’s RDF foundation (Section 3). The use of `rdf:Property` with `rdfs:domain` and `rdfs:range` declarations — rather than OWL’s `owl:ObjectProperty` and `owl:DatatypeProperty` — reflects the deliberate choice of RDF as the simplest, most generic representation for DPP data.

4.2.1. `dpp:hasPrecursor` (The Assembly Link)

- **Type:** `rdf:Property`
- **Domain:** `dpp:ProductPassport`
- **Range:** `dpp:ProductPassport`
- **Purpose:** Links a parent product DPP to the DPPs of its constituent components. It allows a query engine to crawl backwards through the production network recursively.
- **Cardinality:** One parent product may have many precursors; each precursor has exactly one parent product.

4.2.2. `dpp:derivedFrom` (The Transformation Link)

- **Type:** `rdf:Property`
- **Domain:** `dpp:ProductPassport`
- **Range:** `dpp:ProductPassport`
- **Purpose:** Used when a raw material is transformed (e.g., textile yarn into a fabric roll, or raw chemical powders into a battery anode). It establishes the semantic lineage without requiring the parent passport to understand the internal chemistry of the input.

- **Cardinality:** Many materials may derive from one or more source materials.

4.2.3. *dpp:massFraction* (The Quantification Attribute)

- **Type:** `rdf:Property`
- **Range:** `xsd:decimal`
- **Purpose:** Declares exactly how much of the linked component’s weight or mass constitutes the final product. Associated with a blank node or reified relation linking a precursor DPP to its parent.
- **Datatype:** `xsd:decimal` (e.g., 0.42 for 42% mass fraction).

Note: While the DPP model is expressed in RDF / RDFS for maximum simplicity, the properties are compatible with OWL and can be refined to `owl:ObjectProperty` or `owl:DatatypeProperty` if OWL reasoning is required for specific applications.

4.3. RDF / JSON-LD Representation

When serialised, the parent passport remains clean, lightweight, and completely generic. It simply maps its assembly topology. The critical insight is that the parent DPP contains **no component data** — only links. The actual composition, hazard, and provenance data for each component lives in its own DPP, stored behind the portal. When a query engine resolves `dpp:hasPrecursor`, it follows the link *within the portal’s triple-store*, retrieving the component’s data in real time. This design directly addresses the modularity problem raised at the conference: “*once the size is limited to a few dozens, you can still fully understand the overall data model. Once it gets to hundreds or thousands of data points, that is impossible*” [7]. By distributing data across linked individual DPPs rather than accumulating it in a single monolithic record, the model keeps each data unit manageable and independently maintainable.

```
{
  "@context": "https://w3id.org/dpp/v1",
  "@id": "urn:dpp:product:batch-2026-001",
  "@type": "dpp:ProductPassport",
  "dpp:productName": "Lithium-Ion Battery Pack Model X",
  "dpp:hasPrecursor": [
```

```

{
  "@id": "urn:dpp:component:cathode-batch-789",
  "dpp:massFraction": {
    "@value": 0.42,
    "@type": "xsd:decimal"
  }
},
{
  "@id": "urn:dpp:component:anode-batch-456",
  "dpp:massFraction": {
    "@value": 0.28,
    "@type": "xsd:decimal"
  }
},
{
  "@id": "urn:dpp:component:electrolyte-batch-123",
  "dpp:massFraction": {
    "@value": 0.15,
    "@type": "xsd:decimal"
  }
}
],
"dpp:derivedFrom": {
  "@id": "urn:dpp:material:lithium-carbonate-batch-001"
}
}

```

4.4. Each DPP as a Knowledge Graph (Named Graph)

In the portal-backed architecture, each individual DPP is not merely a JSON-LD document but a **knowledge graph** stored as an RDF Named Graph within the portal's triple-store. A Named Graph is an RDF graph identified by an IRI, allowing the portal to manage each DPP as a distinct, addressable unit within a larger graph database. This design has several implications.

First, **independent lifecycle management**: each DPP can be updated, versioned, or deprecated independently of other DPPs. When a component manufacturer corrects a hazard classification, only that component's DPP is

updated; all parent product DPPs that reference it via `dpp:hasPrecursor` automatically reflect the correction upon resolution, without requiring synchronised updates across the supply chain.

Second, **SPARQL queryability**: because each DPP is a named graph within a national portal’s triple-store, each portal can execute cross-DPP SPARQL queries — for example, traversing an entire supply chain from final product to raw materials in a single query, or identifying all products containing a specific substance of concern across all DPPs it hosts.

Third, **access control granularity**: as discussed in Section 5, the portal can enforce ODRL access policies at the level of individual named graphs or sub-graphs, providing branch-level access control that is not possible when data is scattered across distributed REO endpoints. The use of persistent IRIs as graph identifiers, as emphasised in the author’s honorary lecture at DPP4EU 2026 where “*on the lowest level is the IRI— the international resource identifier*” [8], ensures that each DPP remains uniquely addressable and resolvable indefinitely.

4.5. IRI as the Language-Neutral Foundation

Before a DPP can be represented as a knowledge graph, its constituent entities — products, components, materials, substances, processes, organisations — must be unambiguously identified. The architecture adopts the International Resource Identifier (IRI) as the lowest level of identification, as emphasised in the honorary lecture: “*on the lowest level is the IRI— the international resource identifier.*” The use of IRIs resolves a fundamental challenge in the European DPP ecosystem: the multiplicity of natural languages across member states. As Figure 1 illustrates, the same concept (“exit”) expressed in multiple natural languages that map to a single IRI. This principle, applied throughout the DPP architecture, enables cross-lingual, cross-jurisdictional interoperability without requiring a single natural language to dominate the system.

The terminology used in DPP knowledge graphs is taken from ontologies, which provide the semantic vocabulary for the DPP data. The ontologies are themselves identified by persistent IRIs, ensuring that the meaning of DPP data is unambiguously interpretable across the entire ecosystem. This layered approach — IRIs for identification, ontologies for semantics, SHACL for structure — follows the principle stated in the honorary lecture: “*everything must be ontologised*” [8]. The ontologisation objective is an essentially global language that is adjusted to local languages via mechanisms, while the user

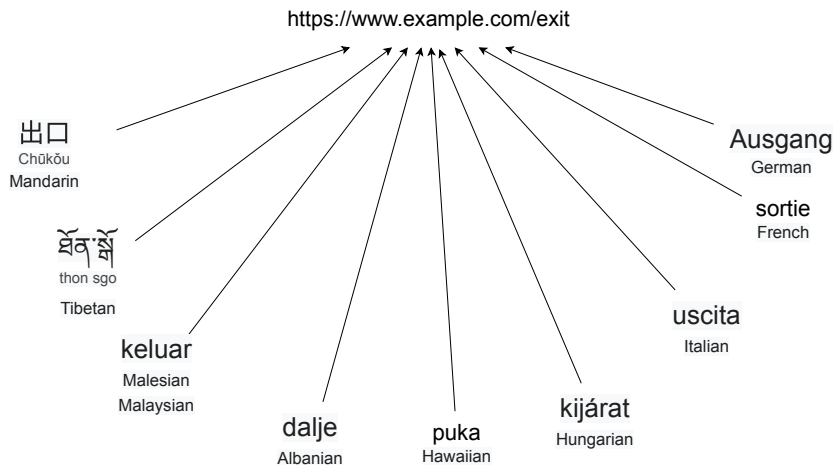


Figure 1: The IRI as a language-neutral identifier. The same concept (“exit”) expressed in multiple natural languages maps to a single International Resource Identifier (IRI). This principle, applied throughout the DPP architecture, enables cross-lingual, cross-jurisdictional interoperability without requiring a single natural language to dominate the system. The terminology is taken from ontologies, which provide the semantic vocabulary for the DPP knowledge graphs.

does not need to be confronted with regulations. The user will need to know his product, but the system handles the regulatory and semantic complexity transparently. This IRI-based foundation directly supports the external entity normalisation stage of the Ingestion Governance Loop (Section 3.2.3): when a DPP references a chemical substance, a standard, or a certificate, the portal controller normalises the reference to a canonical IRI, ensuring consistent identification across all DPPs in the triple-store.

4.6. Production as a Network: The Core Architectural Contrast

A central insight of the proposed architecture is that production is a network, not a linear chain. As stated in the honorary lecture at DPP4EU 2026 [8]: “*production is a network. It’s at the very best case a tree. You have an awful lot of things that are being assembled into something. But it’s definitely not linear*”. This observation has profound architectural consequences. The traditional approach of hardcoding a static supply chain into a single product’s DPP — whether as a flat-file payload or as a blockchain record — fundamentally misrepresents the structure of production. A flat-file DPP embeds

all upstream data in a single document, creating a monolithic record that is stale on creation and impossible to update consistently. A blockchain DPP replicates the supply chain as an append-only ledger, creating unnecessary redundancy and immutability where dynamic updatability is required. Both approaches treat the supply chain as a linear chain; neither captures the reality that production is a network.

The Networked Individual DPP model proposes a fundamentally different mapping: the production network maps to a **DPP graph network**, where each product in the production network has its own DPP, and the DPPs are linked via semantic properties (`dpp:hasPrecursor`, `dpp:derivedFrom`). The supply chain information is not embedded in any single DPP; it is *extracted from the DPP repository by following the linked DPPs* [8]. Figure 2 shows the production network in terms of products and processes, and Figure 3 shows the corresponding DPP graph network.

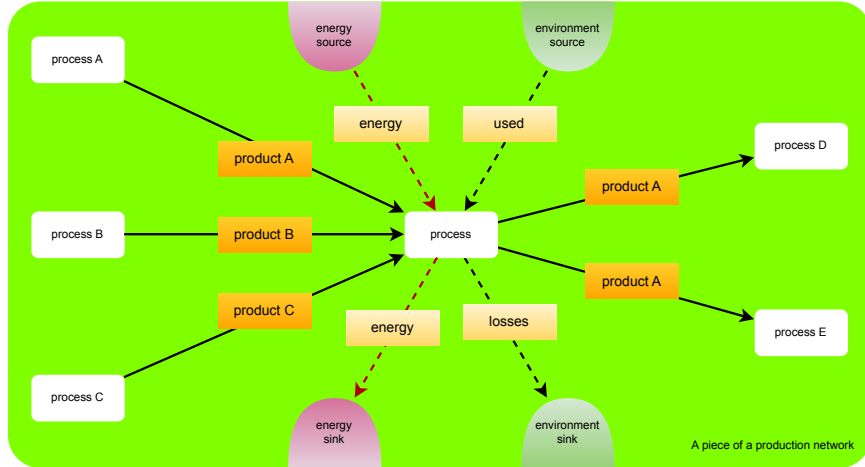


Figure 2: The production network in terms of products and processes. Each process (A–F) produces a product, and products flow between processes as inputs and outputs. The network is a directed acyclic graph, not a linear chain: products are assembled from multiple inputs, and a single product may be used in multiple downstream processes. Each product flow has an associated DPP.

The contrast between the three approaches is summarised in Table 3.

The key architectural distinction is this: in the flat-file and blockchain approaches, the supply chain is *embedded* in the product’s DPP. In the networked graph approach, the supply chain is *emergent* from the links between

Table 3: Comparison of supply chain representation approaches.

	Flat-File DPP	Blockchain DPP	Networked Graph DPP (Proposed)
Supply chain model	Linear chain, embedded in single document	Linear chain, replicated as append-only ledger	Network (DAG), represented as linked graphs
Data structure	Single flat payload per product	Blockchain transaction per product	RDF named graph per product, linked via semantic properties
Update propagation	None — data stale on creation	Append-only — old data preserved, new data added	Instant — single update to component DPP propagates to all parents upon resolution
Data duplication	Full duplication across all parent DPPs	Full replication across all nodes	Zero duplication — links, not data
Query capability	Limited to single DPP payload	Limited to chain traversal	Full SPARQL queries across the entire DPP network
Access control	Binary — full payload or nothing	Binary — on-chain or off-chain	Granular — branch-level ABAC with ODRL policies
IP protection	None — all data exposed in payload	None — all data on ledger	Branch-level filtering — proprietary branches protected
Scalability	Degrades with supply chain depth	Degrades with replication overhead	Scales — each DPP is independently manageable

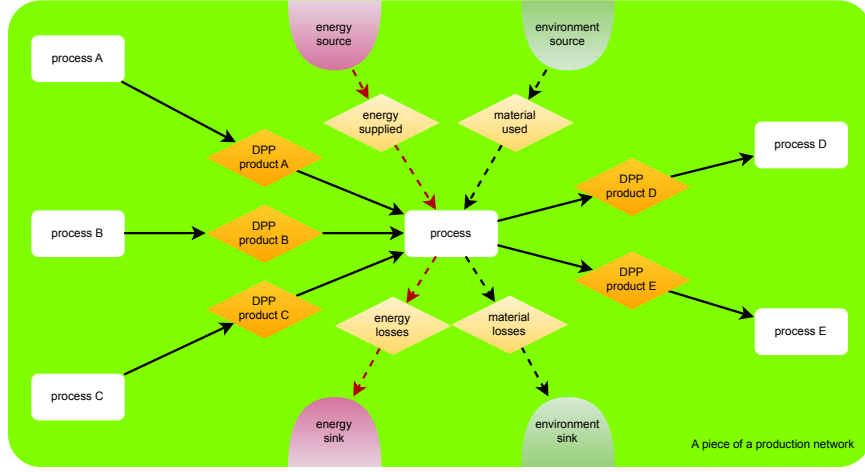


Figure 3: The corresponding DPP graph network. Each product in the production network has its own DPP, stored as a knowledge graph behind the portal. The DPPs are linked via `dpp:hasPrecursor` (assembly links) and `dpp:derivedFrom` (transformation links). The supply chain information is extracted by traversing the linked DPPs within the portal’s triple-store — not by embedding it in a single monolithic record. The portal controller maintains and updates the link network as the production network evolves.

individual DPPs. The supply chain does not exist as a stored data structure; it is constructed on demand by traversing the linked DPPs within the portal’s triple-store. This design also enables capabilities that neither the flat-file nor the blockchain approach can support. A SPARQL query can recursively follow the `dpp:hasPrecursor` links, pull data attributes from each child graph, and compile a real-time, mathematically sound Life Cycle Assessment (LCA) [8].

The DPP network also captures energy flows, waste streams, and utilities. As the honorary lecture explained, each stream in the production process — including waste and utilities — can be associated with “*a DPP or something of that kind*”, though it need not be a complete passport; what matters is that it carries “*the relevant information*” [8]. For some flows, such as electricity from the grid, no DPP exists; instead, the LCA links to **secondary data** from external lifecycle inventory databases to capture the impacts of energy production [8]. The combination of linked DPPs for material streams and secondary data for utility flows enables comprehensive system-boundary LCA: because the portal stores both the mass and energy data associated

with each production step, a query engine can traverse the network and aggregate impacts across the entire system boundary [8].

4.7. *Resolving the Link Rot Problem*

Under the portal-backed model, because the individual graph of each component was pushed *behind the portal* at the time of manufacture, the link remains entirely resolvable forever. The query engine follows the link internally within the portal archive, retrieving the necessary data fields smoothly even decades later. This is in stark contrast to the REO-hosted approach, where a component manufacturer going out of business would return a 404 Not Found error when a query engine tries to resolve `dpp:hasPrecursor`. The parent product’s carbon footprint calculation or material circularity calculation would fail because the component data is simply unreachable. The conference discussion on data persistency (see Section 2.1) directly motivates this design. The portal-backed architecture answers this question structurally rather than procedurally: data persists because it is stored in a governed repository, not at an unmanaged endpoint that may disappear at any time.

4.8. *Resolving the Data Update and Consistency Problem*

The networked model also resolves the data update problem inherent in monolithic DPPs. In the monolithic model, if a component’s data is embedded in 100 parent product DPPs, updating the component data requires updating 100 separate DPPs—a logistically impossible task across organisational boundaries. In the networked model, the component data lives in exactly one place (the component’s own DPP behind the portal), and all 100 parent products reference it by link. A single update to the component DPP propagates instantly to all parent products upon resolution. This design also addresses the concern raised at the conference about data correctness and dynamic updates: “*how we are going to solve this issue or to guarantee the sustainability and working of*” the data over time [8]. The architectural answer is that the portal maintains a single authoritative copy of each DPP, with versioning and provenance metadata, and all references resolve to the current authoritative version.

5. Portal Architecture and Controller Governance

5.1. *The Portal as Semantic Proxy Archive*

The central architectural proposal of this paper is that all DPP data—including product data originally provided by the Responsible Economic Op-

erator (REO) — is stored and maintained *behind a federated portal*, rather than at distributed REO endpoints. The portal acts as a **semantic proxy archive**: a federation of per-member-state triple-stores — one per EU member state — that collectively ingest, validate, store, and serve DPP data as RDF named graphs. Each national instance behaves identically in terms of API, data model, and access governance, while operating on the data relevant to economic operators registered in that jurisdiction. Figure 4 provides a conceptual overview of the full architecture, showing the relationships between the federated repository, the ontologised regulations and SHACL brick library, the federated controller and portal, and the tailored app generation pipeline.

This design places the responsibility for sustainable, long-term data custody with a clearly identifiable entity per jurisdiction — the national portal operator — rather than dispersing it across hundreds of thousands of REOs whose operational continuity cannot be guaranteed over the decades-long lifespan envisioned for DPP data. Each national portal instance is not merely a registry that maps identifiers to external endpoints; it is the authoritative repository of the DPP data for economic operators registered in that jurisdiction. The EC central registry, whose implementing act was identified by the European Commission representative at DPP4EU 2026 as the first piece of legislation to be adopted [7], provides the identifier resolution and regulatory framework, while the portal provides the data storage and governance layer. The two are complementary: the registry points to the portal, and the portal serves the data.

5.2. Federation of National Portals

The federated portal model envisions one portal per EU member state, each behaving identically in terms of API, data model, and access governance, but operating on data relevant to economic operators registered in that jurisdiction. Thus, the portal is not a single central server: it is a federation of interoperable national portals, each storing and governing the DPP data of economic operators registered in its jurisdiction. This federation addresses sovereignty concerns while maintaining architectural uniformity. A common portal specification provides a unified architectural baseline that proactively addresses the emergence of parallel approaches across jurisdictions, as called for by a JTC 24 conference participant.

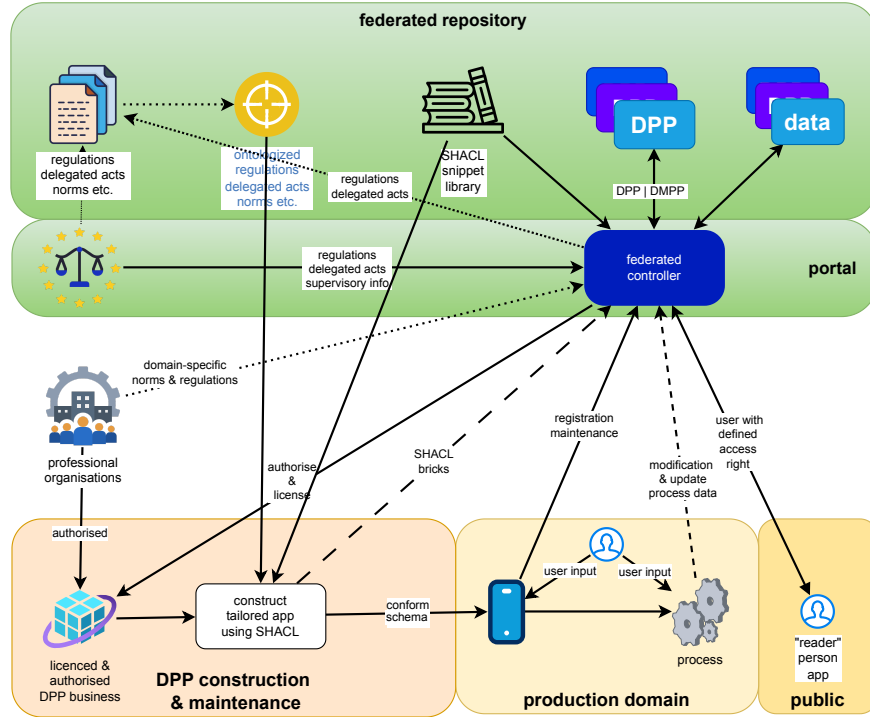


Figure 4: Conceptual overview of the portal-backed DPP architecture. The federated repository (top) stores DPP data, ontologised regulations, delegated acts, norms, and supervisory information. Professional organisations authorise and license SHACL bricks, which are assembled into schemas for tailored app generation. The federated controller and portal (centre) govern access, modification rights, and updates. Licensed DPP businesses construct tailored applications from the SHACL schemas, ensuring that user-entered data conforms to regulations by construction. The “reader” person app provides public access to DPP data.

5.3. The Ingestion Governance Loop

When an REO submits DPP data to the portal, the data passes through an **Ingestion Governance Loop** — a sequence of validation, normalisation, and storage operations performed by the portal controller before the data is committed to the triple-store. This loop ensures that all ingested DPPs are semantically valid, cryptographically authenticated, and normalised to canonical ontological entities before they become queryable.

5.3.1. SHACL Validation

The first stage of ingestion is validation against SHACL shapes. SHACL (Shapes Constraint Language) is a W3C standard for describing and validating RDF graphs against a set of structural constraints called “shapes.” The portal controller maintains a library of SHACL shapes — one set for each product category and applicable regulation — and validates every submitted DPP against the shapes corresponding to its product type and regulatory context. If validation fails, the DPP is rejected with a structured error report indicating which shapes were violated and which data points failed. This design directly addresses the conference observation that machine-readable product data must become checkable by machine-readable regulation (see Section 6.2 for a detailed discussion of SHACL conformity checking). The portal-backed architecture makes this a present capability by hosting both the DPP data and the SHACL validation shapes in a unified triple-store within each national portal instance, enabling automated compliance checking at ingestion time.

5.3.2. Cryptographic Alignment

Each submitted DPP must be cryptographically signed by the REO that authored it. The portal controller verifies the signature against the REO’s registered public key, which is bound to the REO’s identity through the trust tier matrix (Section 5). This ensures that only authenticated, authorised economic operators can submit DPP data to the portal. The cryptographic signature also provides non-repudiation: once a DPP is submitted and signed, the REO cannot deny authorship of the data. While the conference discussion on data carrier verification focused on verifying the physical carrier (e.g., QR code on a product) [7], the portal-backed architecture extends verification to the data layer itself: every DPP in the triple-store is cryptographically authenticated.

5.3.3. External Entity Normalisation

DPP data frequently references external entities: chemical substances (identified by CAS numbers or EC numbers), standards (identified by standard numbers), certifiers (identified by accreditation body identifiers), and materials (identified by various classification schemes). The portal controller normalises these references to canonical IRIs using a set of external registry lookup services. For example, a reference to a chemical substance submitted as “CAS 7440-44-0” is normalised to a canonical IRI such as `urn:echa:substance:7440-44-0`, linking it to the ECHA database entry for that substance. This normalisation ensures that all DPPs in the triple-store use consistent, resolvable identifiers for external entities, enabling cross-DPP queries (e.g., “find all products containing this substance”) to execute correctly. This design is grounded in the conference observation that the DPP must link to standards, measurements, and certificates through the quality infrastructure [7].

5.4. The Resolution Governance Loop

When a user (consumer, professional, circular operator, or authority) requests DPP data from the portal, the request passes through a **Resolution Governance Loop** — a sequence of authentication, authorisation, graph slicing, and serialisation operations performed by the portal controller before the data is returned to the requester. This loop ensures that each user sees only the data branches they are authorised to access.

5.4.1. Authentication and Authorisation

The requester is first authenticated via the EC Authorisation App (Section 5.6), which leverages eIDAS 2.0 digital identity wallets to establish the requester’s identity. The requester’s attributes (user category, jurisdiction, organisational affiliation, certifications) are then evaluated against the ODRL access policies attached to each branch of the requested DPP. This step implements the ABAC framework described in Section 5. The conference observation from the battery passport context — that the actor should see only “*those data that he’s allowed to see but nothing more*” [8] — is precisely the behaviour implemented by this stage.

5.4.2. Graph Slicing via Dynamic SPARQL CONSTRUCT

Once the requester’s access rights are determined, the portal controller constructs a **filtered sub-graph** of the requested DPP — a view that con-

tains only the data branches the requester is authorised to see. This is implemented as a dynamic SPARQL `CONSTRUCT` query: the controller generates a query that traverses the DPP’s named graph, selects only the triples whose attached ODRL policies permit access for the requester’s attributes, and constructs a new RDF graph containing only the authorised triples. The resulting filtered graph is then serialised in the requested format (JSON-LD, Turtle, RDF/ XML) and returned to the requester. This approach — dynamic, query-time filtering rather than pre-computed views — ensures that access policies can be updated without regenerating stored views, and that the most current policy is always applied.

5.4.3. Link Traversal Within the Portal

When the filtered DPP is returned to the requester, it may contain `dpp:hasPrecursor` and `dpp:derivedFrom` links to other DPPs within the portal. The requester may then request those linked DPPs, which pass through the same Resolution Governance Loop. Critically, because all linked DPPs reside within the same portal triple-store, link traversal never produces 404 `Not Found` errors (unless a DPP has been explicitly revoked). The link rot problem — identified in Section 2 as a systemic risk of the distributed model — is structurally eliminated. This architectural design directly answers the data persistency concern raised in Section 2.1: data persists because it is stored in a governed repository, not at an unmanaged endpoint.

5.5. Relationship to the EC Central Registry

The portal-backed architecture is complementary to, not competitive with, the EC central registry. The registry provides:

- **Identifier resolution:** mapping product unique identifiers (as defined in EN 18215 [1]) to portal endpoints.
- **Regulatory metadata:** hosting the SHACL shape library and ontologised regulations (Section 6).
- **Public key infrastructure:** registering REO public keys for cryptographic verification.
- **Audit and compliance:** logging portal operations for regulatory oversight.

The portal provides:

- **Data storage:** hosting DPP named graphs in a per-member-state triple-store.
- **Ingestion governance:** SHACL validation, cryptographic alignment, entity normalisation.
- **Resolution governance:** ABAC-based graph slicing, SPARQL CONSTRUCT filtering.
- **Link traversal:** resolving `dpp:hasPrecursor` and `dpp:derivedFrom` within the portal.

This division of responsibility resolves the tension observed at DPP4EU 2026, where the Materials Commons initiative acknowledged that it does not “*run servers in the end*” [8]. The portal-backed architecture fills this gap: it runs the servers, stores the data, and governs access, while the EC central registry provides the identifier, regulatory, and oversight layer.

5.6. The Public Repository for Ontologised Regulations and Standards

A critical component of the portal architecture is a **public repository for ontologised regulations and standards**, hosted on the EC DPP registry alongside the SHACL shape library. This repository contains machine-readable representations of legal requirements, delegated acts, and harmonised standards that apply to each product category. The conference principle that “*everything needs to be ontologized*” [8] is realised through this repository: regulations, standards, data models, and validation shapes are all represented as semantic, machine-readable artefacts within a unified framework. Section 6 discusses the public repository, the SHACL brick library, and the two-stage design toolchain in detail.

5.7. The Portal API

The portal exposes a standardised REST API that enables economic operators and application developers to interact with the portal controller programmatically. As a conference participant emphasised at DPP4EU 2026 [7], the API should cover “*all major business cases of registering... first of creating a digital product passport, being able to update it being able to delete or revoke it and then finally to register it with the European Commission’s*

DPP registry”. A standardised API is essential because, as another participant observed [7], “*the implementation of an API creates no unique selling point for anyone*” and leaving API implementation to small teams is “*already too much for them*”. The portal API aligns with the JTC 24 API module (EN 18222 [5]) and the Asset Administration Shell API specifications [7], which are “*already in review and publicly accessible*” [8].

The portal API is organised around five operation groups corresponding to the DPP lifecycle:

1. **Authentication:** The requester presents an eIDAS 2.0 digital identity credential to the EC Authorisation App (Section 5.6), which returns a short-lived bearer token containing the requester’s attributes (user category, organisation, jurisdiction, certifications).
2. **DPP Lifecycle Management:** Authenticated REOs can create, read, update, revoke, and register DPPs. Each operation passes through the Ingestion Governance Loop (Section 4.2) for SHACL validation, cryptographic alignment, and external entity normalisation before the DPP is committed to the triple-store.
3. **DPP Resolution:** Authenticated users can request a DPP by its unique identifier. The request passes through the Resolution Governance Loop (Section 4.3), which evaluates ODRL policies against the requester’s attributes and returns a filtered sub-graph containing only the authorised branches.
4. **Link Traversal:** Authenticated users can request linked DPPs by following `dpp:hasPrecursor` and `dpp:derivedFrom` links returned in a resolved DPP. Each linked DPP request passes through the same Resolution Governance Loop.
5. **SPARQL Query:** Advanced users (professional users, circular operators, market surveillance authorities) can submit SPARQL queries against the portal’s triple-store. The portal controller executes the query with dynamic ODRL-based graph filtering, returning only triples from branches the requester is authorised to access.

The full API specification, including endpoint paths, HTTP methods, request and response schemas, and authentication flows, is provided in Appendix [Appendix A](#). Example interaction sequences covering the five operation groups are provided in Appendix [Appendix B](#).

6. ABAC Framework and ODRL Access Policies

6.1. *The Need for Granular Access Control*

The ESPR mandates that DPPs serve multiple user categories with different information needs and access rights. A consumer scanning a QR code on a product needs access to public-facing information about composition, repairability, and environmental impact. A professional user — a repair technician, a product designer, or a supply chain partner — needs access to technical specifications and component data. A circular economy operator — a recycler or a remanufacturer — needs access to detailed material composition and hazard data. A market surveillance authority needs access to compliance documentation and full supply chain provenance. The Responsible Economic Operator (REO) who authored the DPP needs access to the complete data record, including proprietary formulation details. The distributed flat-file model does not inherently support this granularity: an REO endpoint either serves the complete DPP payload or a subset, but without a formal framework for attribute-based, branch-level filtering. As a representative from the battery passport context described at the DPP4EU 2026 conference (see Section 1.4), the requirement is that the actor is authenticated and authorised so that they see only the data they are permitted to access — nothing more. This is precisely the behaviour that the Attribute-Based Access Control (ABAC) framework presented in this section implements, following the NIST Attribute-Based Access Control definition [19] adapted to the DPP context.

6.2. *Five Standardised User Categories*

We define five standardised user categories within the DPP access governance framework, each with a distinct set of attributes, permissions, and authentication requirements.

6.2.1. *dpp:PublicUser*

The public user category encompasses consumers and any individual accessing DPP data without formal organisational affiliation. Access is limited to public-facing data branches: product name, basic composition summary, environmental impact indicators, repairability information, and recycling instructions. Authentication is lightweight — the EC Authorisation App verifies the request origin but does not require organisational identity. No sensitive commercial data, supply chain details, or proprietary formulations are exposed to this category.

6.2.2. *dpp:ProfessionalUser*

The professional user category encompasses repair technicians, product designers, supply chain partners, and other economic operators with a legitimate professional interest in the product’s DPP data. Access includes technical specifications, component identifiers, supplier information (at the organisational level), and compliance documentation. Authentication requires organisational identity verification through the trust tier matrix (Section 5.5). Access is granted on a need-to-know basis [7], reflecting the conference observation that professional users “*will never get the full access or they shouldn’t get the full access*” to all data in the DPP [7].

6.2.3. *dpp:CircularOperator*

The circular operator category encompasses recyclers, remanufacturers, and waste management operators who need detailed material composition, hazard classifications, and disassembly information to safely and effectively recover materials from end-of-life products. Access includes full composition data (including substances of concern), hazard statements, and disassembly instructions. Authentication requires both organisational identity and competence verification — the operator must demonstrate that it is a legitimate circular economy actor with the appropriate certifications and authorisations for handling the materials in question. This category directly addresses the data persistency concern discussed in Section 2.1. The portal-backed architecture ensures that circular operators can access this data decades after manufacture, provided their credentials remain valid.

6.2.4. *dpp:MarketSurveillanceAuthority*

The market surveillance authority category encompasses national regulatory bodies responsible for verifying product compliance with applicable regulations. Access includes the full DPP record: all data branches, all supply chain provenance, all compliance documentation, and all REO declarations. Authentication requires governmental identity verification through eIDAS 2.0 public sector identity credentials. This category has the broadest access rights, reflecting the regulatory mandate of market surveillance authorities under the ESPR. Full access is granted to authenticated regulatory bodies, and only to them, answering the conference question about who should have access to the full information [7].

6.2.5. *dpp:InternalDataOriginator*

The internal data originator category encompasses the REO that authored and submitted the DPP. Access includes the complete DPP record plus internal metadata: submission history, validation results, cryptographic signatures, and provenance logs. This category is unique in that it includes both the public and restricted data branches that the REO authored, plus internal management data. Authentication requires the REO's registered organisational identity and cryptographic key, bound through the trust tier matrix. This category enables the REO to manage its own DPPs: update data, correct errors, submit new versions, and monitor access logs.

6.3. *ODRL Branch-Level Access Policies*

The ABAC framework evaluates access decisions based on attributes of three entities: the **subject** (the requester, characterized by user category, organisational identity, jurisdiction, and certifications), the **object** (the DPP data branch, characterized by its classification, sensitivity level, regulatory context, and applicable policies), and the **environment** (the request context, including time, jurisdiction, and active legal exceptions). This model follows the NIST Attribute-Based Access Control definition [19], adapted to the DPP context. Access policies are encoded using the W3C Open Digital Rights Language (ODRL) [28], attached to individual branches within each DPP's RDF named graph. Each data branch in a DPP carries one or more ODRL policies that specify which user categories may access it, under what conditions, and for what purposes. For example, a public-facing composition summary branch carries an ODRL policy permitting access to all user categories. A proprietary formulation branch carries an ODRL policy permitting access only to

`dpp:InternalDataOriginator` and

`dpp:MarketSurveillanceAuthority`.

A supplier identity branch carries an ODRL policy permitting access to

`dpp:ProfessionalUser`,

`dpp:CircularOperator`,

`dpp:MarketSurveillanceAuthority`, and

`dpp:InternalDataOriginator`, but not to `dpp:PublicUser`.

This branch-level policy attachment enables fine-grained access control that is not possible with binary, endpoint-level access models, directly addressing the conference concern about transferring confidential data only to authorised parties [7].

6.3.1. Example ODRL Policy

The following ODRL policy, expressed in JSON-LD, attaches to a proprietary formulation branch within a DPP, restricting access to the internal data originator and market surveillance authorities:

```
{
  "@context": [
    "https://w3id.org/dpp/v1",
    "https://www.w3.org/ns/odrl.jsonld"
  ],
  "@type": "odrl:Policy",
  "odrl:permission": [
    {
      "odrl:action": "odrl:read",
      "odrl:target":
        "urn:dpp:product:batch-2026-001#formulation",
      "odrl:assignee": [
        {"@id": "dpp:InternalDataOriginator"},
        {"@id": "dpp:MarketSurveillanceAuthority"}
      ],
      "odrl:constraint": {
        "odrl:leftOperand": "dpp:regulatoryContext",
        "odrl:operator": "odrl:eq",
        "odrl:rightOperand": "ESPR"
      }
    }
  ]
}
```

This policy states that the `odrl:read` action on the formulation branch is permitted for `dpp:InternalDataOriginator` and `dpp:MarketSurveillanceAuthority` subjects, provided the regulatory context is `ESPR`. The portal controller evaluates this policy during the Resolution Governance Loop (Section 4.3) when a requester attempts to access this branch.

6.4. *Dynamic SPARQL CONSTRUCT Filtering*

The Resolution Governance Loop implements the ABAC framework through dynamic SPARQL CONSTRUCT queries. When a requester accesses a DPP, the portal controller:

1. Authenticates the requester via the EC Authorisation App (Section 5.6).
2. Retrieves the requester’s attributes (user category, organisation, jurisdiction, certifications).
3. Retrieves the DPP’s named graph from the triple-store.
4. Evaluates the ODRL policies attached to each data branch against the requester’s attributes.
5. Constructs a filtered sub-graph containing only the branches for which the ODRL evaluation returns `Permit`.
6. Serialises and returns the filtered graph to the requester.

This dynamic, query-time filtering ensures that access policies are always evaluated against the most current ODRL policies and the most current requester attributes. If a policy is updated (e.g., a new regulation expands circular operator access to additional data branches), the update takes effect immediately for all subsequent requests without requiring regeneration of stored views. The SPARQL CONSTRUCT mechanism is well suited to this task because it operates natively on RDF graphs and can express arbitrarily complex filtering conditions over both data and policy metadata.

6.5. *The Trust Tier Matrix*

Before a user can be assigned to one of the five categories and granted access to DPP data, their identity and qualifications must be verified through a three-pillar trust tier matrix.

6.5.1. *Pillar 1: Legal Existence Verification*

The first pillar verifies that the organisation requesting access is a legally registered economic operator. This is performed through lookup in the European Economic Operators’ Registration and Identification (EORI) system [15] and the VAT Information Exchange System (VIES) [16]. The EORI number uniquely identifies an economic operator within the EU customs and tax system, while VIES confirms VAT registration.

For a `dpp:MarketSurveillanceAuthority`, legal existence is verified through the governmental identity registry of the member state.

For a `dpp:PublicUser`, this pillar is not required (no organisational identity needed).

6.5.2. Pillar 2: Competence Vetting

The second pillar verifies that the organisation has the competence and certifications required for its claimed user category.

For a `dpp:CircularOperator`, this includes waste management permits, recycling certifications, and hazardous material handling authorisations.

For a `dpp:ProfessionalUser`, this includes professional qualifications relevant to the product category (e.g., electronics repair certification, automotive technician license). Competence is verified through W3C Verifiable Credentials [29] issued by accredited certification bodies. The verifiable credentials are cryptographically signed by the issuer, stored in the requester’s eIDAS 2.0 digital identity wallet, and presented to the portal controller during authentication. This pillar realises the quality infrastructure community’s emphasis on linking DPPs to standards, measurements, and certificates (see Section 7.4): certifications are verifiable credentials, and their provenance is traceable to the issuing accreditation body.

6.5.3. Pillar 3: Identity Binding via eIDAS 2.0

The third pillar binds the organisational identity (Pillar 1) and the competence credentials (Pillar 2) to a verified natural person acting on behalf of the organisation. This is performed through eIDAS 2.0 [17], the European Digital Identity Framework, which provides a legally binding digital identity infrastructure across all EU member states. The requester presents an eIDAS 2.0 digital identity credential that links their natural person identity to their organisational role, and the portal controller verifies the binding.

For a `dpp:MarketSurveillanceAuthority`, the eIDAS 2.0 credential confirms governmental authority.

For a `dpp:InternalDataOriginator`, the eIDAS 2.0 credential confirms that the individual is authorised to submit DPP data on behalf of the REO. This three-pillar trust model ensures that access is granted only to verified organisations, with verified competence, acting through verified individuals — a model that directly addresses the conference concern about data correctness and what is needed to prove that a data point is correct [7].

6.6. The EC Authorisation App

The authentication and authorisation pipeline is managed by an EC Authorisation App, which serves as the centralised identity gatekeeper for the DPP system. The app leverages eIDAS 2.0 digital identity wallets to establish requester identity, retrieves organisational identity and competence

credentials from the requester’s wallet, and evaluates them against the trust tier matrix. Upon successful verification, the app issues a short-lived authorisation token containing the requester’s attributes (user category, organisation, jurisdiction, certifications), which the portal controller uses to evaluate ODRL policies during the Resolution Governance Loop. This centralised model ensures uniform authentication across all member state portals, consistent evaluation of trust tier requirements, and a single point of policy enforcement. The conference concern about data carrier verification and phishing [7] is addressed at the data access layer: only authenticated, authorised requesters can access DPP data, and impersonation attempts are structurally prevented by the eIDAS 2.0 identity binding.

6.7. Resolving the IP Leakage Risk

The ABAC framework with ODRL branch-level policies directly resolves the intellectual property leakage risk identified in Section 2.4. In the distributed flat-file model, an REO endpoint serves the complete DPP payload to any requester who reaches it, with no granular filtering. In the portal-backed model, the Resolution Governance Loop constructs a filtered sub-graph for each requester based on their attributes and the ODRL policies attached to each branch. Proprietary formulation data, supplier identities, and process parameters are protected by ODRL policies that restrict access to authorised user categories. A competitor who authenticates as a `dpp:ProfessionalUser` or `dpp:PublicUser` will see only the public and professional branches; the proprietary branches are filtered out by the SPARQL `CONSTRUCT` query. The conference observation that “*this data really can be shared and if there’s a compromise between confidentiality and transparency*” [7] is resolved by the ABAC framework: the compromise is not a binary choice but a granular, policy-driven decision at the branch level. Each data branch can be independently classified, independently governed, and independently exposed or protected, based on the regulatory requirements and the commercial sensitivity of the data it contains. The transfer of intellectual property rights, identified as an open issue at the conference — “*there’s a transfer of IPI. This is not covered yet in the standards because it’s not quite clear how to organise this*” [7] — can be modelled within this framework through ODRL policies that specify IPI-specific access conditions, updated dynamically as IP rights transfer between organisations.

7. SHACL as a Unified Data Modelling Language

7.1. *The Dual Role of SHACL*

A central innovation of the proposed architecture is the use of SHACL (Shapes Constraint Language) [27] not merely as a validation tool but as a **unified data modelling language** that serves two complementary purposes: (1) as the basis for conformity checking of submitted DPP data against applicable regulations and standards, and (2) as the generative source for user interfaces — DPP websites and application program interfaces — via SHACL-to-HTML rendering engines. This dual role transforms SHACL from a passive validation layer into an active modelling substrate that drives both data ingestion and data presentation.

At DPP4EU 2026, SHACL was described by a conference participant as “*a very elegant approach*” for DPP validation, and a NILU researcher was introduced as speaking about “*SHACL information models and how they will be solving all our problems*” [7]. The conference discussion revealed that the community recognises SHACL’s potential but has not yet connected it to a portal-backed architecture where SHACL shapes serve as the single source of truth for both validation and interface generation. The proposed architecture makes this connection explicit: SHACL shapes are stored in the public repository on the EC DPP registry, used by the portal controller for ingestion validation (Section 4.2.1), and used by application developers to generate user interfaces that, by construction, ensure user-entered data conforms to regulations and standards.

7.2. *SHACL for Conformity Checking*

The first role of SHACL is conformity checking: verifying that submitted DPP data satisfies the structural, cardinality, datatype, and semantic constraints defined by applicable regulations and harmonised standards. As a JTC 24 representative observed at DPP4EU 2026, machine-readable product data must become checkable by machine-readable regulation [7]. The portal-backed architecture makes this a present capability by maintaining SHACL shapes — one set for each product category and applicable regulation — in the same semantic framework as the DPP data itself.

When an REO submits DPP data to the portal, the Ingestion Governance Loop validates the data against the SHACL shapes corresponding to the product’s category and regulatory context (Section 4.2.1). If validation fails, the DPP is rejected with a structured error report indicating which shapes

were violated and which data points failed. This design ensures that only regulation-compliant data enters the triple-store. The conformity checking is not a post-hoc audit but a precondition for ingestion: data that does not conform to the applicable SHACL shapes is never committed to the portal.

The conference observation that “*you still have to look in the book whether what you see on the screen is actually according to requirements*” [7] is resolved structurally: the “book” (the regulation) is represented as SHACL shapes, and the checking is performed automatically at ingestion time. The quality infrastructure community’s emphasis on linking DPPs to standards, measurements, and certificates (see Section 7.4) is realised through SHACL shapes that reference the applicable standards and certificate schemas, ensuring that the DPP data is not only structurally valid but also semantically aligned with the regulatory requirements it claims to satisfy.

7.3. SHACL for GUI Generation via DASH

The second role of SHACL is interface generation: using SHACL shapes as the basis for generating user interfaces — web forms, mobile app screens, and API schemas — that enable economic operators to enter DPP data. This is implemented through SHACL-to-HTML rendering engines such as DASH (SHACL Application Dashboard) [6] and the open-source `shacl-form` library [26] developed by the Universitäts- und Landesbibliothek Darmstadt. At DPP4EU 2026, a participant from the DigiPass project described how SHACL can “*validate a knowledge graph that is coming as ingested information and it can also be used as has been done in our project to automatically generate a form for manual data ingest*” [8].

This dual-use capability is the key insight: the same SHACL shapes that validate submitted data at the portal can also generate the data entry forms that produce the data in the first place. When an application developer builds a DPP data entry tool — a web platform or a mobile app — they use the SHACL shapes from the public repository to generate the user interface. The generated interface presents fields corresponding to the SHACL properties, enforces cardinality and datatype constraints, and validates user input against the shapes in real time. By construction, the data entered through such an interface conforms to the regulations and standards encoded in the SHACL shapes. This eliminates the gap between data entry and data validation: the same shapes govern both, ensuring that data is “born compliant” rather than being checked for compliance after the fact.

7.4. *The SHACL Brick Library*

To enable modular, maintainable, and regulation-aware interface generation, we propose a **SHACL brick library**: a collection of modular, regulation-compliant SHACL shape components — “bricks” — that are reviewed against applicable regulations, delegated acts, and harmonised standards, and then assembled into complete SHACL schemas for specific product categories and regulatory contexts.

A SHACL brick represents a coherent unit of regulatory or standard-based requirements: for example, a “chemical composition brick” that encodes the data points and constraints required by REACH for substance declaration, a “carbon footprint brick” that encodes the data points required by the ESPR for environmental impact declaration, or a “battery composition brick” that encodes the data points required by the Battery Regulation. Each brick is independently developed, independently validated against the source regulation, and independently versioned. When a new regulation is adopted, or an existing regulation is amended, only the affected bricks are updated; all schemas that do not use those bricks remain unchanged.

The brick library is hosted on the public repository for ontologised regulations and standards (Section 6.7), making it accessible to all DPP stakeholders: portal operators, application developers, economic operators, and market surveillance authorities. The conference principle that “*everything needs to be ontologised*” (Section 3.5) is realised through the brick library: regulations, standards, data models, and validation shapes are all represented as modular semantic artefacts within a unified framework.

7.5. *The Two-Stage SHACL Design Toolchain*

The generation of complete SHACL schemas from the brick library requires a **two-stage SHACL design toolchain**: a brick generator and a schema assembler.

7.5.1. *Stage 1: The Brick Generator*

The brick generator is a tool that takes ontologised regulatory and standard-based information as input and produces individual SHACL bricks as output. The input is domain-specific semantic information: ontologies, vocabularies, and structured representations of regulatory requirements for a specific product domain (e.g., batteries, textiles, construction products, chemicals). The output is a set of SHACL shape components, each encoding a coherent unit of

regulatory requirements. The brick generator requires that semantic information for product-domain-specific regulations and standards be generated and made available (Section 6.8). This is a non-trivial task: it requires domain experts (regulatory specialists, standards experts, product category specialists) to collaborate with ontology engineers to produce machine-readable representations of regulatory requirements. The conference discussion on building “*ontologies with experts*” [8] directly motivates this stage: the brick generator is the tool that bridges the gap between domain expert knowledge and machine-readable SHACL shapes.

7.5.2. Stage 2: The Schema Assembler

The schema assembler is a tool that takes a set of SHACL bricks and assembles them into a complete SHACL schema for a specific product category and regulatory context. The assembler selects the relevant bricks based on the product category, the applicable regulations, and the applicable harmonised standards, and combines them into a unified SHACL shape graph. The assembled schema is then used for two purposes: (1) by the portal controller for ingestion validation, and (2) by application developers for interface generation via DASH or shacl-form. The assembler ensures that the assembled schema is consistent — no conflicting constraints, no duplicate properties, no missing mandatory shapes — and that it covers all regulatory requirements applicable to the target product category. When a regulation is updated, the assembler re-generates the schema from the updated bricks, ensuring that all downstream applications — portal validation, data entry forms, API schemas — reflect the most current regulatory requirements.

7.6. Licensed DPP Application Ecosystem

The SHACL-driven architecture creates a new category of economic actor: **licensed DPP application providers** — software businesses that specialise in building, maintaining, and distributing regulation-compliant DPP generation and maintenance applications for specific product domains. For large companies, the DPP task may be handled as part of their internal digitalisation programme; for smaller businesses, however, generating a DPP can be very costly in terms of the time required to learn about the regulations and delegated acts, and the process of registering and maintenance [13]. Licensed DPP application providers fill this gap by offering expertly designed software tools that run on small computer systems, including PCs and mobile phones,

handling the full DPP lifecycle — creation, registration, update, and maintenance — without requiring the economic operator to develop deep regulatory or semantic expertise. As the DigiPass consortium envisioned, the approved SHACL snippet library serves as a very valuable source for generating such applications [13].

7.6.1. The Three-Stage Licensing Pipeline

The generation of licensed DPP applications extends the two-stage SHACL design toolchain (Section 6.5) with a third stage: **app generation, review, and licensing**. The full pipeline is:

1. **Brick generation** (Stage 1): domain experts and ontology engineers produce SHACL bricks from ontologised regulations and domain-specific semantic information.
2. **Schema assembly** (Stage 2): the schema assembler combines bricks into complete SHACL schemas for specific product categories and regulatory contexts.
3. **App generation and licensing** (Stage 3): licensed DPP application providers construct tailored applications from the assembled SHACL schemas, using SHACL-to-HTML rendering engines such as DASH or shacl-form. The resulting application is reviewed, approved, and licensed for distribution.

7.6.2. Dual Authorisation

The licensing of DPP application providers follows a dual authorisation model, reflecting the dual nature of the regulatory and professional requirements. First, the EC, through the portal authority, authorises the software business to operate as a licensed DPP application provider, verifying its legal existence and operational competence. Second, professional organisations (e.g., industry associations, standardisation bodies) authorise the provider’s domain-specific competence, ensuring that the applications it produces are technically and regulatorily appropriate for the product categories they cover. This dual authorisation model mirrors the trust tier matrix (Section 5.5): legal existence is verified through the first pillar, and domain competence is verified through the second pillar, but applied to software providers rather than data requesters.

7.6.3. The Review Board

A **review board** — a technical body established at the authority/portal level — validates submitted SHACL bricks, assembled schemas, and generated applications against the applicable regulations, delegated acts, and norms [13]. The review board has three responsibilities:

1. **Brick review:** each SHACL brick submitted to the public library is reviewed for regulatory accuracy, semantic correctness, and consistency with the source regulation.
2. **Schema review:** assembled SHACL schemas are reviewed for completeness (all applicable regulatory requirements covered), consistency (no conflicting constraints), and correctness (no missing mandatory shapes).
3. **Application review:** generated DPP applications are reviewed for regulatory compliance, correct use of the portal API, and foolproofness (see Section 6.6.4).

The review board is funded through a dedicated EU budget line that finances the maintenance of the open-access standards repository, the SHACL brick review process, and the federation-service APIs [13].

7.6.4. Foolproof by Construction

The defining property of a licensed DPP application is that it is **fool-proof by construction**: the application generates RDF graphs that, when submitted to the portal, will not fail SHACL validation. This guarantee is structural, not accidental. Because the application’s data entry forms are generated from the same SHACL shapes that the portal controller uses for ingestion validation (Section 4.2.1), the forms enforce exactly the constraints that the portal will check. Cardinality, datatype, and semantic constraints are validated in real time during data entry; the user cannot submit a DPP that violates a shape constraint because the form itself prevents it. As a conference participant from the DigiPass project described, SHACL can “*validate a knowledge graph that is coming as ingested information and it can also be used as has been done in our project to automatically generate a form for manual data ingest*” [8]. The licensed DPP application ecosystem operationalises this dual-use capability as a guarantee: the same shapes that validate at the portal generate the forms at the point of data entry, ensuring that data is “born compliant” (Section 6.3).

7.6.5. Portal API Integration

Licensed DPP applications must integrate with the portal API (Section 4.6) for all DPP lifecycle operations: authentication, DPP creation, update, revocation, registration with the EC central registry, link traversal, and SPARQL query. This requirement ensures that:

- All DPP data generated by licensed applications passes through the Ingestion Governance Loop, receiving SHACL validation, cryptographic alignment, and external entity normalisation.
- All DPP updates propagate correctly through the linked DPP network via `dpp:hasPrecursor` and `dpp:derivedFrom`.
- All access requests pass through the Resolution Governance Loop, with ABAC policies enforced uniformly regardless of the originating application.

The portal API thus serves as the single integration point between licensed applications and the portal infrastructure, ensuring architectural consistency across the entire application ecosystem.

7.7. The Public Repository for Ontologised Regulations and Standards

The brick library and the assembled SHACL schemas are hosted in a **public repository for ontologised regulations and standards**, maintained on the EC DPP registry alongside the central identifier resolution service. This repository contains:

- **Ontologised regulations:** machine-readable representations of the ESPR, the Battery Regulation, REACH, and product-specific delegated acts, expressed as RDF graphs with links to the original legal text.
- **Ontologised standards:** machine-readable representations of harmonised standards (EN 18215–18232 [1]) and sector-specific standards, expressed as RDF graphs.
- **SHACL bricks:** modular shape components encoding regulatory and standard-based requirements, produced by the brick generator.

- **Assembled SHACL schemas:** complete shape graphs for specific product categories and regulatory contexts, produced by the schema assembler.
- **Domain ontologies:** product-domain-specific ontologies (e.g., battery ontology, textile ontology, construction ontology) that provide the semantic vocabulary for the SHACL shapes.

The repository is public, freely accessible, and continuously updated as regulations evolve and new product categories come under DPP requirements. The conference aspiration that standards should be “*publicly accessible, and they should be for free, they should not be sold by DIN and other organisations*” [8] is operationalised through this repository: ontologised regulations and standards are hosted on a public EC infrastructure, accessible to all DPP stakeholders without payment. Both the regulations (as ontologised legal text) and the validation shapes (as SHACL bricks and schemas) are available in machine-readable form, enabling automated compliance checking at scale.

7.8. Domain-Specific Semantic Information for Brick Generation

The brick generator (Stage 1 of the toolchain) requires that semantic information for product-domain-specific regulations and standards be generated and made available. This is the foundational layer of the SHACL architecture: without domain-specific ontologies and structured regulatory representations, the brick generator has no input, and no SHACL bricks can be produced. As the Materials Commons presentation at DPP4EU 2026 described, this involves “*the implementation of the semantic framework network and the infrastructure setup*” [8].

The domain-specific semantic information must cover:

- **Regulatory requirements:** the specific data points, thresholds, and constraints mandated by the ESPR, delegated acts, and product-specific regulations for each product category.
- **Standard-based requirements:** the specific data points, measurement methods, and conformity assessment procedures defined by harmonised standards (EN 18215–18232) [1, 2, 3, 4, 5] and sector-specific standards.

- **Data models:** the structured representations of product, component, and material data for each product category, including properties, relationships, and cardinalities.
- **Vocabularies:** the controlled vocabularies, code lists, and classification schemes used in each product domain (e.g., material classification codes, hazard categories, recycling codes).

This domain-specific semantic information is the input to the brick generator, which produces SHACL bricks that encode these requirements in a machine-actionable form. The bricks are then assembled into complete schemas by the schema assembler and published in the public repository, where they are available for portal validation and interface generation.

7.9. The SHACL-Driven Data Lifecycle

The SHACL-centric architecture creates a coherent data lifecycle that spans from regulation to data entry to validation to storage to access:

1. **Regulation ontologisation:** regulations and standards are represented as machine-readable semantic artefacts in the public repository.
2. **Brick generation:** the brick generator produces SHACL bricks from the ontologised regulations and domain-specific semantic information.
3. **Schema assembly:** the schema assembler combines bricks into complete SHACL schemas for specific product categories.
4. **Application generation and licensing:** licensed DPP application providers construct tailored applications from the assembled schemas, reviewed and approved by the review board (Section 6.6).
5. **Data entry:** economic operators enter DPP data through the licensed applications, which validate input against the SHACL shapes in real time.
6. **Ingestion validation:** the portal controller validates submitted DPP data against the same SHACL shapes during the Ingestion Governance Loop.
7. **Storage:** validated DPP data is committed to the portal’s triple-store as RDF named graphs.
8. **Access:** the Resolution Governance Loop filters and serves DPP data based on ABAC policies (Section 5).

At every stage, the SHACL shapes serve as the single source of truth for what constitutes valid, regulation-compliant DPP data, making the conference aspirations for machine-checkable regulation and universal ontologisation operational within a single, coherent architectural framework.

8. Chemical Data Governance and Cross-Registry Linking

8.1. The Chemical Data Challenge

A particularly challenging category of DPP data is chemical composition and hazard information. The ESPR, REACH, the Battery Regulation, and product-specific delegated acts all require that DPPs carry data on substances of concern, hazardous classifications, and material composition. However, this data category sits at the intersection of three structural tensions: (1) the regulatory mandate for transparency, (2) the commercial sensitivity of proprietary formulations, and (3) the scientific authority of established chemical registries. A representative from the materials industry captured this tension directly at the DPP4EU 2026 conference [8], noting that the DPP must address “*material safety, our chemical safety processes*” while also acknowledging that there is “*a lot more that we need from this community in order to meet us in the middle*”. The distributed flat-file model resolves none of these tensions: it either exposes proprietary data or withholds it, and it either duplicates chemical hazard information from authoritative sources or omits it entirely.

8.2. The Duplication Problem

Chemical hazard data — substance identities, hazard classifications, SVHC listings, safety data sheet content — is already maintained by authoritative public databases. The European Chemicals Agency (ECHA) [14] maintains the authoritative registry of substances registered under REACH, including SVHC candidate list entries, harmonised classifications under CLP, and authorisation list entries. PubChem [25] maintained by the U.S. National Library of Medicine provides open-access chemical substance information including structures, properties, and links to regulatory entries. Duplicating this data within individual DPPs creates three problems. First, **data drift**: when ECHA updates a hazard classification or adds a substance to the SVHC candidate list, all DPPs that have duplicated the old classification become stale and inconsistent. Second, **scale**: with potentially millions of DPPs across the European single market, each carrying duplicated chemical data,

the storage and update burden becomes prohibitive. Third, **authority**: a hazard classification stated in a DPP carries less authority than a classification linked to the ECHA database, because the DPP data is self-declared by the REO while the ECHA data is regulatorily authoritative.

8.3. Cross-Registry Linking via External Entity Normalisation

The portal-backed architecture resolves the duplication problem through the external entity normalisation stage of the Ingestion Governance Loop (Section 4.2.3). When an REO submits a DPP containing chemical substance references, the portal controller normalises these references to canonical IRIs using external registry lookup services (e.g., “CAS 7440-44-0” to `urn:echa:substance:7440-44-0`). The DPP stores the *link*, not the *data*. When a user accesses the DPP through the Resolution Governance Loop, the portal can resolve the link in real time, retrieving the current authoritative hazard classification from ECHA and presenting it alongside the REO’s composition declaration. This approach ensures that: (1) the DPP always reflects the most current authoritative hazard data, (2) the storage burden is minimised (only links, not full data sets), and (3) the authority of the data is preserved (it comes from ECHA, not from the REO’s self-declaration). The conference discussion on material declaration reinforced this principle, with a participant advocating for “single point of definition” (SPOD) — “*one repository where we say that’s how we identify a material and that are the definitions we serve to tell you what are the hazard statements what is the granularity*” [7]. The cross-registry linking approach operationalises the SPOD principle within the portal-backed architecture.

8.4. The Quality Infrastructure Connection

The cross-registry linking strategy extends beyond chemical databases to the broader quality infrastructure: standards, measurements, and certificates. As a representative of the German quality infrastructure emphasised at the DPP4EU 2026 conference [8], the quality infrastructure “*allows to link the data from the DPP to exactly those standards, measurements and certificates*”, with the further observation that certificate data could be directly implemented in the DPP and that it is important to trace what standards were used to assess the information [8]. The portal-backed architecture realises this vision through external entity normalisation: when a DPP references a standard (e.g., “EN 18215”), a measurement (e.g., a specific test method), or a certificate (e.g., a conformity assessment certificate from an accredited

body), the portal controller normalises these references to canonical IRIs linking to the relevant quality infrastructure databases. The conference discussion also highlighted the importance of the quality infrastructure for trust: “*I think we have many DPP proposals trying to make verifiable data without addressing, in particular, the QI, the quality infrastructure, and I think it’s a waste of time because this is what we need to have in order to trust the DPP structures at all*” [8]. The three-pillar trust tier matrix (Section 5.5) complements this by ensuring that the organisations providing certified data are themselves verified through EORI / VIES, W3C Verifiable Credentials, and eIDAS 2.0 identity binding.

8.5. *The SVHC and CLP Alignment*

A specific challenge is the representation of Substances of Very High Concern (SVHC) and harmonised classifications under the Classification, Labelling and Packaging (CLP) regulation in DPPs. A conference participant from the materials manufacturing perspective noted at DPP4EU 2026 [7] that the industry needs consistency: “*if we talk about SVHCs, follow the same definition from CLP and the candidate list from ECHA whether you work in textiles, automotive tyres, or furniture*”. The cross-registry linking approach ensures this consistency structurally: rather than each REO defining SVHC and CLP classifications independently in their DPPs, all DPPs link to the same ECHA-maintained candidate list and CLP harmonised classification entries. When ECHA adds a substance to the SVHC candidate list, all DPPs containing that substance automatically reflect the updated status upon resolution, without requiring any action from the REOs. This design directly addresses the conference concern about consistency and eliminates the risk of divergent SVHC classifications across product categories.

8.6. *The Industry Dilemma: Proprietary Composition and Partial Disclosure*

A well-documented tension in DPP implementation is the resistance of chemical and materials companies to disclosing proprietary composition data through a system that could expose it to competitors. The conference discussion on confidential data sharing [7] exemplifies this concern. The ABAC framework with ODRL branch-level policies (Section 5) provides the architectural answer: proprietary composition branches are protected by ODRL policies that restrict access to

`dpp:InternalDataOriginator` and
`dpp:MarketSurveillanceAuthority`,

while public-facing composition summaries (which may omit proprietary details) are accessible to `dpp:PublicUser`. The cross-registry linking strategy adds a further dimension: for hazard classification data that is already public in ECHA, the DPP can link to ECHA rather than duplicating the classification, allowing the public to see the hazard information without the REO disclosing its proprietary formulation. The REO declares the presence of the substance (and links to its ECHA entry) without disclosing the exact quantity or formulation context. This partial disclosure model — link to public hazard data, protect proprietary composition — resolves the Industry dilemma within a single architectural framework. The conference observation that “*this data really can be shared and if there’s a compromise between confidentiality and transparency*” [7] is realised through this granular, branch-level approach.

8.7. Interoperability with Industry Data Platforms

The cross-registry linking strategy must also address interoperability with industry-specific data platforms that already maintain chemical and materials data. A conference presentation demonstrated interoperability between ChemX and ContainerX at the Hannover Messe, where “*SECA as a client picked up the data from Covestro that was polymer data*” [8]. The portal-backed architecture supports such interoperability through external entity normalisation: when a DPP references a material from an industry platform (e.g., a Covestro polymer grade), the portal controller normalises the reference to a canonical IRI that links to the industry platform’s entry for that material. This design accommodates both public registries (ECHA, PubChem) and industry platforms (ChemX, Materials Commons), treating them uniformly as external authoritative sources that DPPs link to rather than duplicate. The Materials Commons initiative (Section 2.2) serves as one such external source. The portal-backed architecture complements Materials Commons by providing the storage, access governance, and ingestion validation that Materials Commons explicitly does not offer.

8.8. The Open-Source Material Characterisation Database

A conference participant described at DPP4EU 2026 [7] the development of “*an open-source database with the material characterisation for everybody to have access to identifiers for free and also to test material compositions*”. This initiative aligns with the cross-registry linking strategy: open-access

material characterisation databases can serve as additional external authoritative sources that DPPs link to. The portal controller’s external entity normalisation can accommodate multiple registries — ECHA for regulatory hazard data, PubChem for chemical substance information, open-source material characterisation databases for materials property data — normalising all references to canonical IRIs and resolving them in real time during the Resolution Governance Loop. This multi-registry approach ensures that DPPs carry the most authoritative, most current, and most comprehensive data available, without duplicating any of it.

9. Discussion

9.1. Beyond Compliance: DPP as Business Enabler

A recurring theme at DPP4EU 2026 was the need to ensure that DPPs are perceived and implemented as more than a regulatory compliance burden. As a conference speaker emphasised at DPP4EU 2026 [9], the goal is to “*make sure that the implementation and the perception of DPPs stay beyond compliance*”. The portal-backed architecture supports this transition: by hosting DPPs as queryable knowledge graphs in a federation of national triple-stores, the portal enables not only compliance checking but also advanced use cases — supply chain analytics, circular economy business model innovation, AI-driven material recovery optimisation, and cross-product substance tracking — that go far beyond the minimum regulatory requirements. The conference observation that DPPs should “*transit from statistic documentation to the intelligent decision support system*” [9] is directly enabled by the portal’s SPARQL-queryable knowledge graph architecture, which transforms static DPP records into a living, queryable substrate for circular economy intelligence.

9.2. The Governance Challenge: Human Agreement, Not Technology

A striking post-conference reflection from the DigiPass CSA project crystallised a conclusion that emerged across the event: the DPP challenge is fundamentally one of governance and human agreement, not technology. As Peter Klein, the DigiPass CSA project leader, wrote in the post-conference summary [13]:

“All the excellent presentations, posters, and podium discussions at the DPP4EU crystallized one particular conclusion in my mind:

the solution of the DPP challenge is more related to human agreement, commitment, and collaboration than to any technical implementation issue.”

This observation was echoed by Paul Breiner of the TRUSTex project, who described the conference’s value as “*the transdisciplinary exchange of technical, regulatory, governance and industry-related perspectives around a common objective: unlocking the Digital Product Passport’s potential to drive the systemic transition towards a Circular Economy*” [13]. The portal-backed architecture proposed in this paper is fundamentally a governance architecture: it provides the structural framework within which human agreement, commitment, and collaboration can be operationalised. The portal is not a technological endpoint; it is the institutional infrastructure that makes cross-organisational data governance possible at scale.

9.3. The Role of DigiPass Europe as Community Sustainer

The conference revealed a clear aspiration within the DigiPass consortium to sustain the community beyond the CSA’s lifetime. As a speaker noted at DPP4EU 2026 [8], the plan is to “*go from DigiPass to DigiPass Europe*” in order to “*create a community around these topics that sustain the effort that has been made so far with everyone*”. The proposed portal-backed architecture aligns with this aspiration: DigiPass Europe could serve as the organisational entity that operates or oversees the federated portal infrastructure, the public repository for ontologised regulations, and the SHACL brick library. The conference observation that “*we need actually for this to act a kind of central hub connecting the industry the stakeholders*” [8] maps directly onto the portal’s role as the semantic and governance hub of the DPP ecosystem. The COST action model, described as a bottom-up initiative open to all interested participants [8], provides a complementary funding and governance mechanism for the ongoing community work, particularly for the domain-specific semantic information generation that the SHACL brick library requires.

The conference format itself — a single-track, “one-for-all” programme deliberately designed to avoid community fragmentation — reflects the same architectural principle underlying the portal-backed model. As Heinz Adolf Preisig observed in the post-conference summary: “*One could argue for separate sessions and longer talks, but it would split the community. Given the*

status of the DPP development affair, I felt a need to see the complete picture and minimise the formation of specialised groups” [13]. The architectural parallel is direct: just as the conference community resisted fragmentation through a unified

10. Conclusion and Roadmap

10.1. Summary of Contributions

This paper has proposed a portal-backed, graph-based, network-structured DPP architecture built on semantic web standards that addresses the systemic vulnerabilities of the current distributed, flat-file model. The architecture comprises six integrated components: (1) the Networked Individual DPP model with semantic linking properties (`dpp:hasPrecursor`, `dpp:derivedFrom`), in which each DPP is a knowledge graph node in a production network; (2) the portal as semantic proxy archive, storing all DPP data — including REO product data — behind a federation of national portals, one per EU member state; (3) an ABAC framework with five standardised user categories and ODRL branch-level access policies, ensuring strictly controlled, granular access for protection of sensitive data; (4) a three-pillar trust tier matrix combining EORI / VIES, W3C Verifiable Credentials, and eIDAS 2.0; (5) SHACL as a unified data modelling language serving both conformity checking and GUI generation, supported by a SHACL brick library, a two-stage design toolchain, a licensed DPP application ecosystem with dual authorisation and review board, and a public repository for ontologised regulations and standards; and (6) a cross-registry linking strategy for chemical data governance, connecting DPPs to authoritative public databases rather than duplicating data. Evidence from the DPP4EU 2026 conference — comprising 150 in-person participants, over 1,700 online participants, and representatives of four European Commission Directorate-Generals, HaDEA, and CEN/CENELEC JTC 24 [13] — demonstrates that the constituent elements of this architecture are independently recognised as necessary by the practitioner community. This paper integrates them into a coherent whole.

10.2. The Roadmap for the European Commission

The proposed architecture aligns with a vision already anticipated within the DigiPass consortium. An internal consortium deliverable anticipated a future in which “*there is indeed a European DPP Registry with: registration, API access, semantic repository*” — a vision that maps directly onto the

portal-backed architecture proposed here [13]. The European Commission’s funding agency has likewise recognised the need for structural coordination: Giuseppe Giovanni Daquino of HaDEA observed at DPP4EU 2026 that the agency “*clearly identified how these projects are clustered through the DPP connection with SSbD processes and its integration and strategic alignment with the European advanced materials innovation ecosystem*” [13]. The federated portal architecture provides exactly this clustering and integration function at the technical level.

Based on the architecture proposed in this paper, we recommend the following roadmap for the European Commission:

10.2.1. Phase 1 (2026–2027): Foundation

- Adopt the federated portal specification as the architectural baseline for the DPP central registry infrastructure, complementing the identifier resolution function with the portal’s data storage and governance layer.
- Establish the public repository for ontologised regulations and standards on the EC DPP registry, beginning with the ESPR, the Battery Regulation, and REACH.
- Pilot the portal architecture with the first product category (batteries), using the Battery Regulation requirements as the initial SHACL shape library.
- Begin the SHACL brick generation process for the battery domain, engaging domain experts and ontology engineers.
- Establish the review board for SHACL bricks, schemas, and licensed DPP applications, with a dedicated EU budget line for ongoing maintenance.
- Define the dual authorisation framework for licensing DPP application providers, involving both the European Commission and relevant professional organisations.

10.2.2. Phase 2 (2027–2028): Expansion

- Extend the portal architecture to the next product categories (textiles, construction products, electronics), generating domain-specific SHACL brick libraries for each.

- Begin licensing the first DPP application providers for the battery domain, using the SHACL schemas produced in Phase 1.
- Deploy the federated portal across all willing EU member states, with a common specification and interoperability testing.
- Integrate the three-pillar trust tier matrix with the EC Authorisation App, leveraging eIDAS 2.0 digital identity wallets.
- Begin cross-registry linking implementation, starting with ECHA for chemical hazard data and expanding to other quality infrastructure databases.
- Expand the licensed application ecosystem to textiles, construction products, and electronics as domain-specific SHACL brick libraries become available.

10.2.3. Phase 3 (2028–2030): Maturation

- Achieve full federated portal coverage across all EU member states.
- Complete the SHACL brick library for all priority product categories under the ESPR.
- Establish the DigiPass Europe community entity as the ongoing custodian of the SHACL brick library and the domain-specific semantic information, with COST action or equivalent funding.
- Extend the cross-registry linking strategy to industry data platforms (Materials Commons, ChemX, sector-specific platforms).
- Explore extension of the portal specification to non-EU countries for global DPP interoperability through UN/CEFACT and ISO/IEC JTC 5.

10.3. The Path Forward

The conference chair’s closing observation — that “*this year’s conference was very inspiring. It has stronger results. It has clearer contributions, a deeper understanding of what the passport is and what it could yet become*” [9] — captures the moment. The DPP community has reached a point of clarity about what is needed: semantic representations, persistent storage, granular access control, quality infrastructure integration, machine-readable regulation, and a licensed application ecosystem that makes compliance accessible

to enterprises of all sizes. What is missing is the integrated architecture that combines these elements. This paper proposes such an architecture. The path forward requires the European Commission, member states, standards bodies, and the practitioner community to collaborate on its implementation — a collaboration that the DigiPass Europe initiative, the COST action, and the JTC 24/JTC 5 standardisation processes are well positioned to sustain.

acknowledgments

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools for analysing the conference transcripts, the post-conference report, and proofreading.

Appendix A. Portal API Specification

This appendix specifies the portal REST API. All endpoints use HTTPS with TLS 1.3. Request and response bodies are JSON (for management operations) or JSON-LD / Turtle / RDF / XML (for DPP data). All requests except authentication require a valid bearer token in the **Authorization** header.

Appendix A.1. Authentication

Table A.1: Authentication endpoint.

Method	Path	Description
POST	/auth/token	Exchange eIDAS 2.0 credential for a bearer token. Returns a short-lived JWT containing the requester’s attributes.

Request body (JSON):

```
{
  "eidasCredential": "<\eIDAS 2.0 digital identity
    credential>",
  "requestedScopes": ["dpp:read", "dpp:write"]
}
```

Response body (JSON):

```
{
  "accessToken": "<JWT>",
  "tokenType": "Bearer",
  "expiresIn": 3600,
  "attributes": {
    "userCategory": "dpp:ProfessionalUser",
    "organisation": "urn:efri:DE123456789",
    "jurisdiction": "DE",
    "certifications": ["urn:vc:electronics-repair-cert"]
  }
}
```

Appendix A.2. DPP Lifecycle Management

POST /dpps — Request body (JSON-LD):

```
{
  "@context": "https://w3id.org/dpp/v1",
  "@id": "urn:dpp:product:batch-2026-001",
  "@type": "dpp:ProductPassport",
  "dpp:productName": "Lithium-Ion Battery Pack Model X",
  "dpp:hasPrecursor": [...],
  "dpp:derivedFrom": {...},
  "dpp:signature": {
    "algorithm": "RS256",
    "value": "<base64-encoded signature>",
    "certificate": "<X.509 certificate URI>"
  }
}
```

POST /dpps — Response body (JSON):

```
{
```

Table A.2: DPP lifecycle management endpoints.

Method	Path	Description
POST	/dpps	Submit a new DPP. The request body is a signed JSON-LD document. Passes through the Ingestion Governance Loop. Returns the assigned DPP URI and validation report.
GET	/dpps/{dppId}	Retrieve a DPP by its unique identifier. Passes through the Resolution Governance Loop. Returns a filtered JSON-LD sub-graph based on the requester's access rights.
PUT	/dpps/{dppId}	Update an existing DPP. The request body is a signed JSON-LD document with the updated content. Requires <code>dpp:InternalDataOriginator</code> role. Triggers re-validation through the Ingestion Governance Loop.
DELETE	/dpps/{dppId}	Revoke a DPP. The DPP is marked as revoked (not physically deleted; provenance is preserved). Requires <code>dpp:InternalDataOriginator</code> role.
POST	/dpps/{dppId}/register	Register a DPP with the EC central registry. Maps the DPP URI to the product unique identifier in the EC registry. Requires <code>dpp:InternalDataOriginator</code> role.

```

    "dppId": "urn:dpp:product:batch-2026-001",
    "status": "accepted",
    "validationReport": {
      "shaclConformant": true,
      "cryptographicVerified": true,
      "entitiesNormalized": 14,
      "errors": []
    }
  }
}

```

GET /dpps/{dppId} — Response body (JSON-LD, filtered):

```

{
  "@context": "https://w3id.org/dpp/v1",

```

```

"@id": "urn:dpp:product:batch-2026-001",
"@type": "dpp:ProductPassport",
"dpp:productName": "Lithium-Ion Battery Pack Model X",
"dpp:hasPrecursor": [
  {"@id": "urn:dpp:component:cathode-b789"},
  {"@id": "urn:dpp:component:anode-b456"}
],
"dpp:filteredBranches": ["dpp:public", "dpp:professional"]
}

```

Note: The `dpp:filteredBranches` property is a metadata annotation added by the portal controller to indicate which branches were included in the response, enabling the client to understand what was filtered out.

Appendix A.3. Link Traversal

Table A.3: Link traversal endpoint.

Method	Path	Description
GET	/dpps/{dppId}/precursors	Retrieve all precursor DPPs linked via <code>dpp:hasPrecursor</code> . Each linked DPP is resolved through the Resolution Governance Loop and returned as a filtered subgraph.
GET	/dpps/{dppId}/derivations	Retrieve all derivation DPPs linked via <code>dpp:derivedFrom</code> . Each linked DPP is resolved through the Resolution Governance Loop and returned as a filtered sub-graph.
GET	/dpps/{dppId}/supplychain	Recursively traverse the full supply chain by following <code>dpp:hasPrecursor</code> and <code>dpp:derivedFrom</code> links. Returns an aggregated JSON-LD graph containing all accessible linked DPPs, each filtered according to the requester's access rights.

Appendix A.4. SPARQL Query

Request body (JSON):

Table A.4: SPARQL query endpoint.

Method	Path	Description
POST	/sparql	Submit a SPARQL CONSTRUCT or SELECT query. The portal controller executes the query with dynamic ODRL-based graph filtering: only triples from branches the requester is authorised to access are returned. The response is returned in the requested serialisation format (JSON-LD, Turtle, RDF / XML, or SPARQL JSONresults).

```
{
  "query": "CONSTRUCT { ?s ?p ?o } \
    WHERE { ?s ?p ?o . \
      ?s dpp:containsSubstance \
        <urn:echa:substance:7440-44-0> }",
  "format": "application/ld+json"
}
```

Response body (JSON-LD, filtered by access rights):

```
{
  "@context": "https://w3id.org/dpp/v1",
  "@graph": [
    {
      "@id": "urn:dpp:product:batch-2026-001",
      "dpp:containsSubstance":
        { "@id": "urn:echa:substance:7440-44-0" },
      "dpp:productName":
        "Lithium-Ion Battery Pack Model X"
    }
  ]
}
```

Appendix A.5. Error Handling

All error responses use standard HTTP status codes and a consistent JSON error body:

Error response body (JSON):

```
{
```


Table A.5: Error response codes.

Code	Meaning	Description
400	Bad Request	The request body is malformed, or the SHACL validation failed. The response includes a structured validation report.
401	Unauthorised	The bearer token is missing, expired, or invalid.
403	Forbidden	The requester's attributes do not permit access to the requested resource or branch.
404	Not Found	The requested DPP does not exist or has been revoked.
409	Conflict	The DPP already exists (on POST) or has been modified since retrieval (on PUT).
422	Unprocessable Entity	The DPP is semantically valid JSON-LD but fails SHACL validation. The response includes the SHACL validation report.

```

"error": {
  "code": "SHACL_VALIDATION_FAILED",
  "message": "The submitted DPP does not conform \
    to the applicable SHACL shapes.",
  "validationReport": [
    {
      "shape": "dpp:BatteryCompositionShape",
      "property": "dpp:carbonFootprint",
      "constraint": "sh:minCount",
      "value": 0,
      "expected": 1
    }
  ]
}

```

Appendix B. Example API Interaction Sequences

This appendix provides concrete examples of API interactions covering the five operation groups defined in Section 4.6.

Appendix B.1. Example 1: REO Submits a New DPP

An economic operator (EO) submits a new DPP for a battery batch. The interaction proceeds as follows:

1. **Authenticate:** The EO presents its eIDAS 2.0 credential to obtain a bearer token.
2. **Submit DPP:** The EO submits the signed JSON-LD DPP via POST /dpps.
3. **Ingestion:** The portal controller validates the DPP against SHACL shapes, verifies the cryptographic signature, normalises external entity references, and commits the DPP to the triple-store.
4. **Register:** The EO registers the DPP with the EC central registry via POST /dpps/{dppId}/register.

Step 1: Authenticate

```
curl -X POST https://portal.example.eu/auth/token \
  -H "Content-Type: application/json" \
  -d '{"eidasCredential": "...",
      "requestedScopes": ["dpp:write"]}'
```

Response: {"accessToken": "<JWT>", "attributes": {...}}}

Step 2: Submit DPP

```
curl -X POST https://portal.example.eu/dpps \
  -H "Authorization: Bearer <JWT>" \
  -H "Content-Type: application/ld+json" \
  -d @dpp-batch-2026-001.jsonld
```

```
# Response: {"dppId": "urn:dpp:product:batch-2026-001",
#            "status": "accepted",
#            "validationReport": {...}}
```

Step 3: Register with EC central registry

```

curl -X POST \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001/register \
  -H "Authorization: Bearer <JWT>"

# Response: {"registryStatus": "registered",
#           "productIdentifier": \
#           "EN18215:DE:123456789:batch-2026-001"}

```

Appendix B.2. Example 2: Consumer Scans QR Code and Reads Public DPP

A consumer scans a QR code on a product and requests the public-facing DPP data. The interaction proceeds as follows:

1. **Authenticate:** The consumer's app presents a lightweight eIDAS 2.0 credential (no organisational identity required) to obtain a `dpp:PublicUser` bearer token.
2. **Resolve DPP:** The consumer's app requests the DPP via `GET /dpps/{dppId}`.
3. **Filter:** The portal controller evaluates ODRL policies against the `dpp:PublicUser` attributes and returns only the public-facing branches (product name, basic composition summary, environmental impact indicators, repairability information, recycling instructions).

```

# Step 1: Authenticate (lightweight)
curl -X POST https://portal.example.eu/auth/token \
  -H "Content-Type: application/json" \
  -d '{"eidasCredential": "...",
      "requestedScopes": ["dpp:read"]}'

# Response: {"accessToken": "<JWT>",
#           "attributes": {
#           "userCategory": "dpp:PublicUser", ...}}

# Step 2: Resolve DPP
curl -X GET \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001 \

```

```

-H "Authorization: Bearer <JWT>" \
-H "Accept: application/ld+json"

# Response: JSON-LD with only public-facing branches
# (dpp:filteredBranches: ["dpp:publicBranch"])

```

Appendix B.3. Example 3: Circular Operator Traverses Supply Chain

A certified recycler needs the full material composition and disassembly information for a product at end-of-life. The interaction proceeds as follows:

1. **Authenticate:** The recycler presents its eIDAS 2.0 credential (including waste management permits and recycling certifications as W3C Verifiable Credentials) to obtain a `dpp:CircularOperator` bearer token.
2. **Resolve DPP:** The recycler requests the DPP via `GET /dpps/{dppId}`. The portal returns a filtered sub-graph including the full composition data, hazard statements, and disassembly instructions.
3. **Traverse Supply Chain:** The recycler requests the full supply chain via `GET /dpps/{dppId}/supplychain`. The portal recursively follows `dpp:hasPrecursor` links and returns an aggregated JSON-LD graph containing all accessible precursor DPPs, each filtered according to the recycler's access rights.

```

# Step 1: Authenticate
curl -X POST https://portal.example.eu/auth/token \
-H "Content-Type: application/json" \
-d '{
  "eidasCredential": "...",
  "verifiableCredentials": [
    "urn:vc:waste-mgmt-permit-DE-123",
    "urn:vc:recycling-cert-EU-456"
  ],
  "requestedScopes": ["dpp:read"]
}'

# Response: {"accessToken": "<JWT>",
#           "attributes": {

```

```
#           "userCategory": "dpp:CircularOperator",
#           ...}}
```

Step 2: Resolve DPP

```
curl -X GET \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001 \
  -H "Authorization: Bearer <JWT>" \
  -H "Accept: application/ld+json"
```

```
# Response: JSON-LD with composition, hazard,
# and disassembly branches
# (dpp:filteredBranches: ["dpp:public",
# "dpp:composition", "dpp:hazard",
# "dpp:disassembly"])
```

Step 3: Traverse full supply chain

```
curl -X GET \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001/supplychain \
  -H "Authorization: Bearer <JWT>" \
  -H "Accept: application/ld+json"
```

```
# Response: Aggregated JSON-LD graph with all
# accessible precursor DPPs
```

Appendix B.4. Example 4: Market Surveillance Authority Runs Cross-Product SPARQL Query

A market surveillance authority needs to identify all products in the portal that contain a specific substance of concern. The interaction proceeds as follows:

1. **Authenticate:** The authority presents its eIDAS 2.0 governmental credential to obtain a `dpp:MarketSurveillanceAuthority` bearer token.
2. **SPARQL Query:** The authority submits a SPARQL CONSTRUCT query via POST `/sparql` that traverses all DPPs in the portal and identifies those containing the target substance. The portal controller

executes the query with full access rights (no ODRL filtering for market surveillance authorities) and returns the aggregated result.

Step 1: Authenticate

```
curl -X POST https://portal.example.eu/auth/token \
  -H "Content-Type: application/json" \
  -d '{"eidasCredential": "<gov credential>",
      "requestedScopes": ["dpp:read","dpp:sparql"]}'
```

```
# Response: {"accessToken": "<JWT>",
#           "attributes": {
#             "userCategory": "dpp:MarketSurveillance",
#             ...}}
```

Step 2: SPARQL query

```
curl -X POST https://portal.example.eu/sparql \
  -H "Authorization: Bearer <JWT>" \
  -H "Content-Type: application/json" \
  -d '{
    "query": "CONSTRUCT { ?dpp \
      dpp:containsSubstance ?substance . \
      ?dpp dpp:productName ?name } \
    WHERE { ?dpp \
      dpp:containsSubstance \
      <urn:echa:substance:7440-44-0> . \
      ?dpp dpp:productName ?name }",
    "format": "application/ld+json"
  }'
```

Response: JSON-LD graph with all DPPs

containing the target substance

Appendix B.5. Example 5: REO Updates a DPP After a Hazard Reclassification

ECHA reclassifies a substance, adding it to the SVHC candidate list. The REO of a product containing that substance updates the DPP to reflect the new classification. The interaction proceeds as follows:

1. **Authenticate:** The REO presents its eIDAS 2.0 credential to obtain a `dpp:InternalDataOriginator` bearer token.
2. **Retrieve DPP:** The REO retrieves the current DPP via GET `/dpps/{dppId}` (with full access rights as the data originator).
3. **Update DPP:** The REO submits the updated DPP via PUT `/dpps/{dppId}` with the corrected substance reference (now linking to the updated ECHA entry via the normalised IRI). The portal controller re-validates the DPP against SHACL shapes, verifies the signature, and commits the updated version to the triple-store. All parent product DPPs that reference this DPP via `dpp:hasPrecursor` automatically reflect the updated classification upon subsequent resolution.

Step 1: Authenticate

```
curl -X POST https://portal.example.eu/auth/token \
  -H "Content-Type: application/json" \
  -d '{"eidCredential": "...",
      "requestedScopes": ["dpp:read","dpp:write"]}'
```

```
# Response: {"accessToken": "<JWT>",
#           "attributes": {
#             "userCategory": "dpp:InternalDataOriginator",
#             ...}}
```

Step 2: Retrieve current DPP

```
curl -X GET \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001 \
  -H "Authorization: Bearer <JWT>" \
  -H "Accept: application/ld+json"
```

Step 3: Update DPP with corrected substance ref

```
curl -X PUT \
  https://portal.example.eu/dpps/\
urn:dpp:product:batch-2026-001 \
  -H "Authorization: Bearer <JWT>" \
  -H "Content-Type: application/ld+json" \
  -d @dpp-batch-2026-001-updated.jsonld
```

```
# Response: {"dppId": "urn:dpp:product:batch-2026-001",  
#           "status": "updated",  
#           "validationReport": {...},  
#           "version": 2}
```


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